The Egyptian Navy after Muhammad Ali Pasha, 1848-1898

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Abstract

This thesis is a study of the Egyptian Navy in the latter 19th century, a force that has so far escaped dedicated historiographical attention due to its limited on-paper capability. This thesis will argue the contrary: that despite its limited size, the force successfully projected seapower and was a vital component of the state's political, territorial, and military strategy. The main sections of the dissertation will detail how the fleet was able to recover from its decline in the 1840s, maintain its technical and operational competence, and effectively operate across the Black Sea, Mediterranean and Indian Ocean theatres despite being navally overmatched. Through a global, multidisciplinary approach synthesising national, regional, imperial, naval, and strategic scholarship, as well as the use of Arabic, French, Turkish, and Russian sources, a fresh perspective is offered on issues such as the Nationalist-Ottomanist debate, the biopolitical regime of the pre-colonial Egyptian state, the applicability of the "great divergence" to 19th-century naval development, and the importance of secondary empires and navies.

Introduction

The Egyptian Navy is of little avail for war purposes. The late Sultan Abdul Aziz forbade the construction and possession of ironclads, and all that Egypt possesses by way of a marine is a ***small fleet of wooden ships.***

The Times, 1877[1]

The Egyptian Navy in the latter 19th century has thus far been the subject of scant historiographical attention, particularly in English. If the earlier Egyptian Navy under Muhammad Ali was at least the subject of Houghton's single article, the history of the fleet in the half-century that followed has warranted neither dedicated articles nor chapters, let alone any full-length monographs. The causes of this neglect are manifold, but the underlying factor is relatively straightforward: as a secondary navy in the service of a secondary, non-European power, and one that was not autonomously engaged in major naval confrontations in this period, the late 19th-century Navy's strategic significance to the course of broader Egyptian history has been largely dismissed, either by omission or through ungenerous comparisons with contemporary navies or its previous, larger form. The primary aim of this dissertation is to argue the contrary: that despite its limited size, the Egyptian Navy from 1848 to 1898 successfully projected seapower on behalf of the Egyptian state and was a vital element of its strategy in achieving its political, territorial, and military objectives. This call to attention could be expanded to include the many other second-tier non-European navies of the period, though that is beyond the scope of this discussion.

While this work is an institutional history of the Egyptian Navy, and thus one of ships and those who designed, built, crewed, and commanded them, it is also one that employs the framing of seapower, as opposed to the more limited naval power approach common in operational-level naval campaign histories. The vital operating distinction, as observed by Till, is that seapower addresses not just power at sea, but power projected from the sea.[2] While this distinction might seem trite,

within naval history there exists a tendency to focus upon the Western paradigm of combat at sea and the major navies which dominated it, a trend that has largely continued even within the modern cultural, biographical, or technically oriented literature.[3] Yet it was the very elements that comprise power from the sea, but count for little in strictly naval power terms, namely, auxiliary non-combat missions such as transport, presence and patrol, and diplomacy, that defined the Egyptian Navy from the mid-1850s onwards.

The choice to write an account of the Egyptian Navy in this period in the "traditional" military history style of institutional framing should not be taken as a dismissal of its broader cultural and societal significance, often given the shorthand label of 'new military history' or 'war and society'.[4] Instead, it is a necessary reflection both of the paucity of baseline empirical accounts currently available in English on the Egyptian Navy in this period, which ensures any such secondary analysis must first be preceded by the establishment of said empirical baseline, as well as the nature of the accessible primary sources, whose perspectives are largely top-down and offer little granularity as to the lived experience of the rank-and-file.

A work of synthesis is therefore an essential precursor to any further study; such is the disparate nature of the available literature, in which basic information, such as the names of ships and the timing of events, is often confused. This is not to say that this article will not address the wider societal debates within the historiography of Egypt of this period, as they pertain to the Navy, but that, as a naval history, it makes no claim to offering any decisive, generalizable input across disciplines on broader subjects, such as the severity of state control and the Ottoman/Egyptian dichotomy.

That the naval history of Egypt in this period is a part of, but peripheral to and neglected within, multiple fields of historical enquiry is dictated by the shape of the wider political geography of West Asia, North and East Africa, as well as the broader Mediterranean, within which Egypt occupies a critical crossroads position that was only enhanced by the completion of the Suez Canal in 1869. Historiographically, the history of "modern" Egypt is typically periodized to 1805, with the inauguration of Muhammad Ali/Alawiyya dynastic rule following the interruption to Ottoman-Mamluk authority represented by Napoleon's infamous invasion. The contestation over the nature of the rule of Muhammad Ali and his successors, arising from this transition, is a key theme of 19th-century Egyptian naval history and a major driver of academic interest in English-language literature, with three main camps emerging in the 20th century.

The earliest approach taken by scholars like Dodwell was that of "great man" theory, lauding Muhammad Ali as a great oriental reformer and moderniser whose personal ambitions and vision drove the development of the Egyptian state and society in a European direction, much as colonial administrators in India did.[5] This position was subsumed from the late 1970s onwards by a national emphasis, best represented by Arabists like Marsot and Talhami, who saw in political assertions of autonomy and widespread societal/military mobilisation the origins of an Egyptian nation-state.[6] Since the 1990s, this nationalist perspective has been challenged by a countervailing Ottomanist approach, whose foundational works are Toledano's 1990 *State and Society in Mid-Nineteenth-Century Egypt* and Fahmy's 1997 *All the Pasha's Men*, which argued respectively that Egyptian elites self-consciously identified with the broader imperial elite and guarded social distinctions against "Egyptians", and that the raising of military forces to seek autonomy was common Ottoman provincial practice and did not generate much national sentiment among conscripts.[7]

Though modern scholarship on Egypt has largely turned away from the confines of political and military history, the unstated bifurcation of Ottomanist versus Arabist/Nationalist has largely held, with Khedival Egypt either implied to be a constituent part of a wider Ottoman World, or a modern break from the past, fostering either colonial modernity or national emergence.[8] As will be shown below, in this recurring thematic debate, the history of the Egyptian Navy, even in the latter part of the 19th century, provides more support for the Ottomanist than the nationalist thesis, though the two positions are not entirely mutually exclusive.

Beginning in the 1820s, Muhammad Ali's expansionist ambitions towards Greece and the Levant brought Egypt and its navy twice into conflict with the European powers, and within the central grand narrative framing of the Eastern Question. Though a dated and Eurocentric diplomatic concept, the Eastern Question was a common direction from which Egyptian and Eastern Mediterranean naval history was approached in the mid to late 20th century, and remains relevant to studies of European perceptions and strategy.[9] It was through these lenses that the Egyptian participation in the Syrian, Crimean, Cretan and 77-78 Russo-Turkish Wars has conventionally been viewed.[10] Egyptian policymakers in the late 19th century were, moreover, aware of broader European politics concerning the Ottoman Empire and exercised considerable political agency, playing various factions against each other to secure concessions with varying success.

The imperial role of Egypt in the "Scramble for Africa" is one of the best-known aspects of 19th-century Egyptian history, second only to the Napoleonic Invasion, at least to English-speaking popular audiences. This is both a reflection of Egypt's, and especially the Suez Canal's strategic significance to British Imperial Defence,[11] as well as the Victorian hagiography of colonial figures like Charles Gordon and the campaigns in Sudan.[12] Yet within academic literature, the secondary Empire carved out by Muhammad Ali and his successors in Sub-Saharan Africa has largely fallen between disciplines. In Africanist historiography of Sudan and Ethiopia, Khedival Egypt's role is that of a temporary invader preceding a greater European influence; in the former case, partly a result of a general lack of historiographical interest in comparison with other African regions,[13] in the latter case, a reflection of the short-lived and ill-fated nature of Egyptian imperialism in the Horn of Africa.[14] Egypt's maritime expansion along the Red Sea and Indian Ocean has similarly attracted peripheral interest in historiographies of these two regions.[15] The topics of Indian Ocean and Islamic World Slavery, which draw continuous sporadic attention, are also of some significance to Egyptian naval history, mostly through the issue's economic and diplomatic importance with regards to East African regional elites and British representatives, given that, unlike in the Egyptian army, enslaved personnel were not a common source of naval recruitment.[16]

From the mid-19th century onwards, growing British influence in the Eastern Mediterranean, Red Sea and West Asia formed a tightening noose around Egyptian autonomy, which, combined with growing influence within Egypt itself, would ultimately culminate in 40 formal and 70 informal years of British occupation from 1882. This resulted in a considerable body of literature addressing Egyptian history from a colonial approach. The bulk of this scholarship is of limited direct interest to students of the Egyptian Navy, inasmuch as it was not perceived as a potential threat to British interests, and institutionally dismantled at the onset of British rule, but is valuable for both its illumination of regional dynamics, and specifically in the Red Sea and Indian Ocean, how Egyptian maritime expansion was enabled and contested by British Imperial representatives. Postcolonial revisionist literature, such as Mitchell's influential 1988 Foucauldian treatise *Colonising Egypt*, has, moreover, partially retro-projected elements of Western discipline and power onto the earlier period of Egyptian history, arguing that the oppressive social regime implemented by the Khedives was an alien European import. The historiography of Italian colonialism, given the Kingdom of Italy's role

as a territorial competitor and a secondary beneficiary of Egypt's imperial collapse, is also of contextual interest, even if direct overlap was limited.[17]

Another unique genre of relevant work are the Americanist military histories of mid to late 19th-century Egypt, inaugurated in the 1950s with works such as Hesseltine and Wolf's *The Blue and the Gray on the Nile*. While initially focused on sympathetically tracing the memoirs of ex-Union and Confederate mercenaries in Egyptian service, more recent works, such as those of Dunn, whose *Khedive Ismail's Army* serves as the only full-length English monograph on the late 19th-century Egyptian military, offer a comprehensive picture of the era's military campaigns. The loyalty, utility, and reliability of foreign officers and advisors of Egyptian service is a recurring theme throughout this period, for as will be seen, regardless of individual intentions, Western personnel were very much a necessary evil.[18]

On the topic of the Crimean War and the period immediately preceding, Russian and Turkish sources, writing either in contemporary naval works or the modern academy, provide key empirical data on the Egyptian Navy in the Black Sea and beyond, compensating for the tendency of Western European historiography to focus on the campaign's territorial and Anglo-French dimensions. Badem's 2010 *The Ottoman Crimean War*, for instance, is valuable for its comprehensive coverage of Constantinople's strategic decision-making before and during the conflict.[19]

While no dedicated histories of the Egyptian Navy in this period are available, empirical and relative data are available in contemporary and modern surveys of global navies. Relatively good coverage exists up to the end of the age of sail, around 1860, courtesy of global longue durée studies such as Glete's, but the growing number of minor navies and the pace of technological change have precluded a similar systemic study of the ironclad era. Instead, the best accessible information in English takes the form of snapshot overviews in the style of Jane's Fighting Ships, though the first edition of that renowned publication was not published until 1898, after the period of this discussion.[20] For comparative purposes, national and multinational overviews of the Ottoman, Russian, British, Chinese and Japanese Navies are illuminating with regard to contemporary strategic naval development, particularly in non-Western contexts.

Despite the disparate nature of English-language historiographical coverage, this dissertation further draws on three underutilised bodies of primary and secondary literature for empirical information. The most important of these are Egyptian military and universal histories, and the Arabic printed primary sources on which they are largely based. Due to the limited scope of this dissertation and limited accessibility outside regional libraries, this work makes no pretensions to historiographical comprehensiveness in this area; nevertheless, the historical traditions in which the Egyptian writers operated require a brief description. Like the broader intellectual landscape, Egyptian history-writing in the 19th and early 20th century was in flux between amateur and professional, medieval annalistic styles and modern, analytic, "Western" approaches, and between stylised prose and direct, journalistic writing; and like most contemporary Western scholarship, patriotic and Egyptian-nationalist in tone to some extent.[21] Sarhank's comprehensive 3-volume work, *Haqaiq al-Akhbar an Duwal al-Bihar (True Narrations of Maritime Nations)*, particularly valuable for the author's second-generation naval background, seniority, firsthand experience, and extensive access and use of state documents, is representative of a transitional style combining the medieval chronicle tradition and modern analytical methodology.[22] Other works, such as the civil servant and senator Amin Sami's 1916-36 6-volume *Taqwim al-Nil (Almanac of the Nile),* are fully committed to the annalist style and are essentially selected reproductions of primary sources.[23] 20th-century histories like the influential nationalist historian Abd al-Rahman al-Rafi'i's *Asr Ismail*

(The Era of Ismail) and amateur military historian Prince Umar Tusun's 1936 *Al-Jaysh al-Misri fil Harb al-Rusiya al-Marufa bi Harb al-Kirim (The Egyptian Army in the Crimean War)*, meanwhile, reflect a sharpening divide between critical and royalist historiography.[24]

Echoing the chronicle style of Arabic literature is a body of early-20th-century works in French, written by French historians and former naval officers with Egyptian Royal patronage. These works represent the positivist and empiricist epistemological traditions of French historiography, as well as a continuation of a legacy of writing on Egypt that began with the "savants". Though not focused on totalizing social structures as was the Annales school, they nevertheless share a dedication to precise detail and a tendency towards voluminousness: Douin's *Histoire du Règne du Khédive Ismaïl* (1933-1941) dedicates four books of more than 600 pages each to the African Empire (Vol. 3 of 3) alone.[25]

A final source of information on the Egyptian Navy are the contemporary accounts of Western travellers, naval officers, soldiers, bureaucrats, and diplomats, which take the form of published books, journal articles or archival reports. While Orientalist tendencies of generalisation, exoticisation, and moralisation are relatively universal, the qualifications of these writers differed dramatically in terms of both their technical expertise in military affairs and their political and cultural familiarity with the region. The anonymous Russian writer of *'Neskol'ko slov o egipetskom flote' (A few words about the Egyptian Fleet)*, for instance, was a naval officer writing for intelligence purposes, and was both able to communicate with Egyptian personnel (likely in Turkish) on such specifics as map-reading and demonstrably familiar with the past developments of the Egyptian Navy; while at the other end of the spectrum, British travel writers tended to produce theatrical, exaggerated claims when observing Westernized military forces, the lives of locals, and Egyptian politics.[26]

The major empirical shortfall of this dissertation, unfortunately shared by much of the recent scholarship on Egypt, is the absence of direct Egyptian archival material. This is a dual outcome of the temporal and financial realities of the program for which it is written, and 21st-century political and military developments that have resulted in the closure of Egypt's archives, now under the control of the security apparatus, to most scholars since 2013.[27] While the Arabic sources occasionally note the paucity and often-conflicting nature of archival material on the Navy, for now, the selected reproductions in the printed primary sources will have to suffice.

As discussed at the start, this is a piece of military naval history, with the primary aim of establishing that the Egyptian Navy in the latter 19th century was an operationally effective force that, despite its reduced size, was able to complete the strategic objectives to which it was tasked by the successors of Muhammad Ali. To that end, the main body of the dissertation will be divided into five thematic chapters that each address how different limitations were overcome.

The first chapter deals with the rapid expansion of the Navy under Muhammad Ali, the challenges faced by the Navy in the 1840s following the second Syrian War, demonstrating how, within Egypt's international context, the pursuit of raw naval strength was a double-edged sword, and ends with the fleet in a state of disrepair.

The second chapter addresses the more systemic and technical factors that dogged the Navy in the latter 19th century: growing industrial and economic disparities, the transition to powered steamships, and the introduction of novel technologies like torpedoes. It will be shown that, despite these challenges, the Egyptian Navy was able to maintain a meaningful degree of operational effectiveness, despite the mixed results of its initiatives.

The third chapter is focused on naval education and recruitment, and how, despite maintaining the social hierarchies and oppressive conscription practices of the Ottoman and Muhammad Ali eras, the officers and men of the mid-to-late-century navy maintained a reasonable degree of training, professionalism, and effectiveness.

The fourth chapter frames the Egyptian Navy's intervention in the Crimean War through the strategic scope of Ottoman politics and diplomacy, illustrating that despite the unfortunate realities of war, the men and ships of the Egyptian Navy performed respectably at a tactical level, projected seapower onto land, and strategically furthered the economic and military ambitions of the Khedives Abbas I and Said.[28]

The fifth chapter negotiates the Egyptian Navy's role in the renewed African expansion of Khedive Ismail, focusing on how it efficiently exerted seapower in the Red Sea and Indian Ocean littorals and ports along the Sudanese, Eritrean and Somali coasts against regional and European imperial contenders, before eventually, through no fault of its own, the Navy became a victim of broader Egyptian imperial collapse towards its ultimate fate of dismemberment under British administration, though even then it played an essential role in Anglo-Egyptian power projection against the Mahdists.

To avoid confusion over the identity of naval vessels, this dissertation will follow the simplified IJMES-based system of Arabic transliteration used by Houghton for the names of ships, with a preference for Arabic over Ottoman Turkish or European versions of the same name, which will be listed in brackets where available in the literature.[29] For the names of individuals and settlements, the standard English rendering most commonly used in modern literature is followed for ease of cross-referencing, though for places subsequently renamed, such as Constantinople, the contemporary form is used to avoid anachronism.

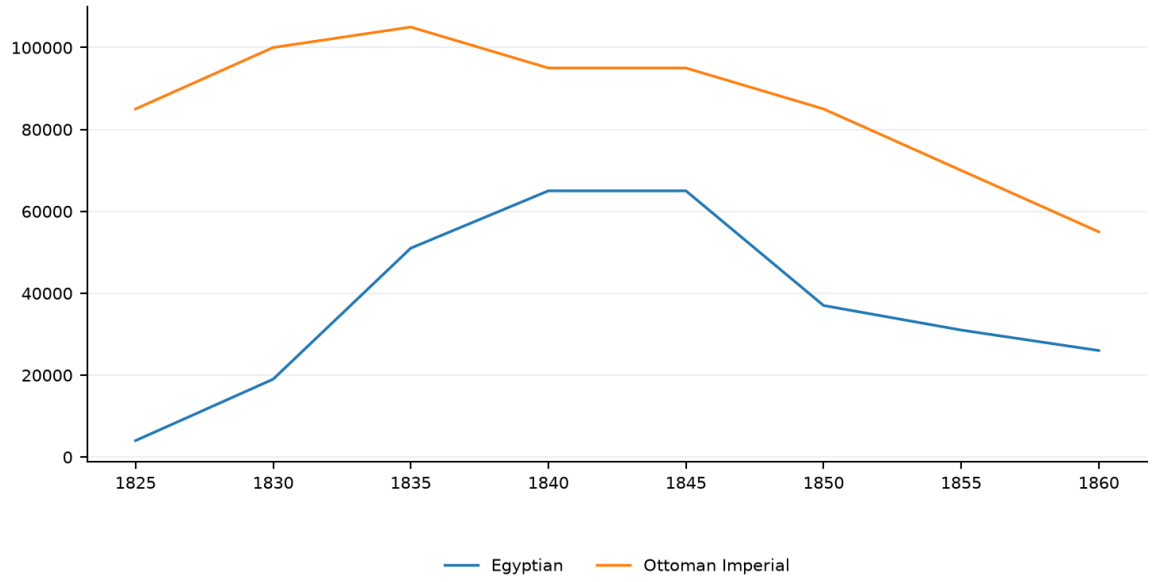
(Paddle) Wheel of Fortune: Countervailing Trends in the 1840s

The history of the Egyptian Navy in the immediate years after 1848 is superficially one of decline. In the singular English monograph on the 19th-century Egyptian fleet, *The Egyptian Navy of Muhammad Ali Pasha*, the golden age of the Egyptian state (and, by extension, its fleet) is dismissed as passing with the death of its founder. The article concludes on a pessimistic note with the fleet rotting at anchor, the English counsel Murray's observation that "all classes whether Turks, or Arabs, not only feel but do not hesitate to say openly that the prosperity of Egypt has died with Muhammad Ali", finishing off with the conservative outlook of his successor, Abbas Hilmi.[30]

To understand Egyptian naval power as it stood at Ibrahim's accession, it is necessary to review its development in the few decades prior, both to contextualise its institutional and technical lineages and to establish a thematic baseline. The Egyptian fleet under Ali was a remarkable force that, as late as 1845, still ranked amongst the world's most powerful. The world's 6th largest at an estimated 65,000 tons, it was overmatched only by the navies of Britain, France, Russia, the Ottoman Empire, and the United States, and unlike them (with the possible exception of the US) was not divided across multiple primary theatres of contention but rather concentrated in the Eastern Mediterranean. To put this displacement in context, the Wali's fleet in 1845 had a displacement roughly double that of Sweden-Norway and Denmark, triple that of Spain or Portugal, and ten times that of the Austrian Empire.[31] In contrast, the fleet in 1850 appeared far less imposing, its total displacement having shrunk by 45% to a mere 37,000 tons through the retirement of half its ships of the line.[32] This decline is even more precipitous in relative terms – in 5-year

intervals from 1840 to 1860, the fleet went in percentage terms from 4.9 to 4.4, then 2.3, 1.8, and 1.3 of global naval displacement, respectively.[33]

On the surface, then, the Egyptian fleet appears to follow Taagepera's famous parabolic concept of Imperial trajectories, which used the territorial and population extents of Empires as shorthand for their relative "power". As the chart below shows, if one were to apply the same comparative logic to Egyptian naval development, one would observe a parabola traced by a rapid expansionist period in the 1820s and 30s, followed by rapid contraction after European intervention in the Second Syrian War.[34]



Total Fleet Displacement (1825-1860)

Figure 1: Total Fleet Displacements (1825-1860) (data from Glete, Navies and Nations, vol. 2, pp. 465–467)

In technical terms, the Egyptian fleet under Muhammad Ali was built to state-of-the-art European designs and incorporated the latest developments in wooden ship construction, matched among contemporary non-Western navies only by the imperial arsenal on the Golden Horn.[35] A case in point was the class of frigates of the *Jeanne D'Arc* type built and outfitted for the Egyptian Navy in the 1820s. Impressed by the visit of the newly commissioned *Jeanne D'Arc* to Alexandria in 1821, Muhammad Ali would order six equivalent vessels from various European shipyards from Venice to Archangel, most of which were launched between 1826 and 1829.[36] In design, these ships represented a French response to the celebrated large frigates used by the British and American navies during the War of 1812, and represented a clear step up in size, firepower and complement over preceding Napoleonic era designs.[37] Nor was this technological advantage limited to foreign-built vessels. The dockyards at Alexandria, under the management of the French naval engineer Louis-Charles Lefébure de Cerisy, could produce first-rate ships of the line with advanced features such as rounded sterns and internal bracing. The largest of these, the 120-gun *Misr* and *Akka,* were equivalent to the contemporary French Improved *Oceans* and as powerful as any warship then afloat.[38]

The rapid expansion of the fleet, combined with the relatively limited training of much of the industrial workforce, did result in certain technical issues. The Egyptian ruler was overcharged for his large, foreign-built frigates, paying 1.7 million francs for a vessel made of inferior materials that

would have been worth only 1.1 million francs even if built of good wood.[39] The earlier first-rates built in Egypt were of relatively poor build quality, suffering from unseasoned timbers, thin copper plating and poor quality control.[40] Perhaps most illustrative of the inexperience of Egyptian dockyards was the case of the large three-deckers *al-Mansura* and *Misr*, which were completed with a draft too deep to leave the shallow harbour of Alexandria, with the latter consequently having to be razeed by a deck.[41] These flaws should not, however, obscure the highly impressive nature of the fleet and its supporting naval infrastructure during the reign of Muhammad Ali, the scale and capabilities of which put Egypt arguably ahead of even the Sultan in Constantinople. If the Egyptian Navy did not usually exceed that of the Porte in quantity, joint operations often showed it to be qualitatively superior. During the 1824 Morean campaign, for instance, a disgusted Ibrahim Pasha wrote that the imperial fleet had to put in at Alexandria as they could not "even fix the masts of their frigates", before snarkily remarking on the amount of food they consumed in Egypt.[42]

Observations on the operational efficiency of the Muhammad Ali-era fleet at sea were mixed. During his reign, the navy saw combat in two main theatres: with the Ottomans in the Aegean against Greek insurgents, and against the Ottomans and their later European backers in the Levant. Western sources tend to dismiss the Egyptian Navy's performance in the former as poor, with observers such as French general Pierre François Xavier Boyer claiming that with only 70–80 armed merchant brigs, the insurgents were "mistress of the sea" and that the Turco-Egyptian fleet was reduced to seeking security in fortified ports for shelter against Greek fireships.[43]

One major reason for this Western dismissal was the 1827 battle of Navarino, in which a powerful Anglo-French-Russian squadron under Codrington inflicted a decisive defeat on a Turkish and Egyptian force at anchor, despite the quantitative superiority of the latter. The specific circumstances of this engagement, however, limit its relevance to evaluating the capabilities of the Egyptian fleet. Western historians tend to accept the claim that poor discipline in the Egyptian fleet led to pistol shots being fired at a British pinnacle, initiating the battle;[44] while the Egyptian view was that the Allies launched a perfidious surprise attack without the customary declaration of war.[45] In any case, there is little doubt that the Egyptian commanders were not seeking action – the squadron commander, Muharrem Bey, not only kept his own ship, the powerful frigate *Murshid Jihad,*[46] initially unengaged, but even sent an officer to parley aboard the *Asia* stating he had no intention to fire despite the British vessel having already engaged the Ottoman flagship.[47]

In contrast with the negative Western view, Egyptian histories provide a reasonably positive appraisal of the fleet's performance during the war in Greece. In its initial Aegean deployment from July 1821 to March 1822, the Egyptian fleet first harried Greek shipping in the Aegean after rendezvousing with its Ottoman counterpart off Ephesus. The combined fleet then moved to anchor at the Ionian port of Zante/Zakynthos, before fighting a series of engagements in the mouth of the Gulf of Preveza/Ambracian Gulf. Thirty Greek ships were claimed sunk, with thirty-six more captured, their crews hung from the masts of their captured vessels in a grisly procession towards the Dardanelles.[48] Indeed, as the war continued, the Egyptian fleet grew more ascendant in its ability to control the sea lanes of communication necessary to sustain the expeditionary force in the Morea through engagements such as that off Cape Spada in 1825, in which Egyptian escorts under Hasan al-Iskandarani defeated the Greek admiral Andreas Miaoulis, sinking two ships and capturing a schooner.[49] It was this reduction of the rebel navy to "little more than a nuisance", combined with the sheer preponderance of force achieved, that enabled the Turco-Egyptian victories on land, but which in turn provided a trigger for the aforementioned Western intervention at Navarino.[50]

While earlier nationalist sources offer post-facto apologies for Muhammad Ali's conscription policies, describing them either as a necessary evil for national unity, going so far as to claim that most conscripts would, through their service, be taught loyalty to the Egyptian nation, there seems to be little doubt that this view was not generally shared by the fellahin on whose behalf it was expressed.[51] If a sense of national solidarity was eventually induced, it was through the shared negative experience of prejudice and mistreatment by the "Turco-Circassian" elite, a process not unanalogous to that which occurred in the late Qing and Austro-Hungarian Empires. In any case, during the early 19th century, the manifestation of these trends into significant political movements remained firmly in the future.

The question of to what end Muhammad Ali ultimately built up his powerful forces is historiographically contentious, falling as established in the introduction between Ottomanist and Nationalist schools of thought. Nationalist writers tended to take the maximalist view that he consciously sought from the start to secure Egyptian independence from central authority and to expand his sphere of influence whenever possible, with Marsot further seeing an institutional state-building aspect in his military expansion.[52] Ottomanists, in contrast, see Ali's goal as narrowly dynastic, given how mid-century elites spoke Turkish and identified with Ottoman values and culture, with the build-up of forces merely to ensure autonomy while retaining Ottoman legitimation, and thus the eventual struggle between Egypt and the Ottomans were "dynastic struggles within the Empire at best, or a civil war at worst".[53]

The debate over the underlying rationale for Egyptian military and territorial expansion, however, does not significantly affect the assessment of the rationale behind Muhammad Ali's Navy, as the operational and tactical requirements to achieve even the minimal aim of dynastic security required the Egyptian fleet to be able to sustain operations besides, and if necessary, against that of the Porte. Thus, regardless of what his ultimate political ambitions may be, the Egyptian fleet would have to remain functionally independent, under effective command and control, and logistically independent.

In strategic terms, the tasks the Egyptian fleet would be tasked with can be divided into three overlapping but escalating tiers, corresponding in analytical terms to varying levels of command of the sea ascribed by naval theorists. That modern naval theory is applicable to the 19th-century Egyptian Navy is in itself instructive, given that the achievement of its three main unstated prerequisites, that of the clear divide between warships and merchantmen, the permanence and institutional independence of the naval branch, and the ability to sustain protracted at-sea operations (notably lacking in galleys), were only fully realised in the Mediterranean around 1800.[54]

In its most basic guise, the Navy should be able to operate lines of communication against opponents in permissive environments and provide auxiliary transport capacity in support of land operations, roughly corresponding with the undisputed, local, and temporary state of naval command described by early 20th-century thinkers like Corbett, or a constabulary navy (rank 8 of 9) in Grove's modern classification system.[55] This tier was arguably achieved as early as 1809 during the campaign against the Wahhabis, for which the Egyptian Wali constructed shipyards in Bulaq near Cairo and in Alexandria to build transports and small brig and corvette-type escorts.[56] The next strategic tier, that of being able to secure said sea lines of communication, roughly corresponds to the local, temporary, but critically prepondering level of command in Corbett's schema, and rank 5 (Adjacent Force Projection Navy) in Grove's 9-tier schema. The Egyptian fleet's ability to achieve this level of sea control against a weaker adversary was tested in the Greek

War and found sufficient, arguably being so successful as to provide a pretext for great power intervention.

The final, most challenging tier was the ability not just to control specific lines/areas of communication, but to deny the same to an opposing naval power of similar if not greater strength; roughly corresponding with a general/permanent/preponderant level of naval command or a Medium Force Projection Navy (rank 4 of 9). Achieving this was an explicit rationale behind Egyptian naval expansion stated in Muhammad Ali's diplomatic communications - a powerful, independent fleet was essential both to prevent Egypt from being dictated to by European powers, as was suffered by Constantinople,[57] and to forestall any potential naval counterattack by the Sultan should war break out.[58] A measure of the strategic weight placed on the fleet can be seen in the fact that in 1830, some 14% of Egypt's total expenditures were spent on the navy, more than double the Ottoman figure of 6 to 7%.[59]

In the 1840s, the Egyptian Navy built up by Muhammad Ali suffered a precipitous decline that can only be attributed to the context of rapidly shifting international developments. The first was the culmination of the Egyptian–Ottoman/Syrian Wars in 1840-41 following great-power intervention, which, at a stroke, inverted the role of the powerful Egyptian fleet from a guarantor to an impediment towards the goal of dynastic independence. Ironically, the Egyptian expansion into Syria was partly driven by a need to secure wood for shipbuilding, as was explicitly emphasised in correspondence between Muhammad Ali and Ibrahim.[60] The first Egyptian–Ottoman war was fought from 1831 to 1833, and resulted in a decisive victory for Muhammad Ali, whose son Ibrahim was granted Syria, Adana and Candia (Crete) with the acquiescence of the great powers at the Peace of Kutahia.[61] The second war was fought from 1839 to 1841, and though militarily it began similarly with one-sided Egyptian victories, the decisive intervention of the great powers despite French protest resulted in an agreement whereby Muhammad Ali lost Syria and other territorial acquisitions gained at the expense of the Porte, but was confirmed in his dynastic hold over Egypt.[62]

The naval events of these wars demonstrate, above all, the porousness of the late Ottoman World in which the Egyptian dynasts operated. The most dramatic incident occurred in July 1839, when a powerful Turkish squadron consisting of 9 ships of the line, 11 frigates, 1 corvette, and 6 smaller vessels defected to the Egyptian side, sailing first to join the Egyptian fleet at Iskenderun, with both fleets then sailing to Alexandria.[63] While the proximate cause of the fleet's defection was the appointment as grand vizier of Husrev Pasha, whose alleged intrigues against its commander Ahmed Fevzi prompted the latter's defection, the fact remains that of the substantial formation, only a brig and a steamer chose to return to Constantinople, the remaining captains presumably accepting the pretext that Husrev planned to surrender the fleet to the Russians.[64] The combined fleet, briefly the most powerful in the Mediterranean, sailed with 12,000 men to Beirut in two squadrons commanded by Fevzi.[65] According to British reports, one was entirely Egyptian and the other composed of 9 Turkish frigates and 1 Turkish Corvette, with only a single Egyptian ship of the line, a split that, while keeping trained formations together, offered little deterrent should the Turkish vessels attempt to re-desert.[66] Though Husrev was removed following repeated insistence by Muhammad Ali,[67] the fleet would only be returned in January 1841 after the war's conclusion under the command of its former British advisor, Baldwin Wake Walker/Yawer Pasha, who had remained loyal to the Porte.[68]

The 1839 incident was not the only high-profile desertion during the Ottoman-Egyptian confrontation. Earlier in 1832, during the first confrontation and following a series of contrived

pursuits with Ottoman warships in which both fleets avoided action, the Egyptian commander Uthman Nur al-Din (Osman Nourredin) deserted in the brig "*Chaaba Gehaat*" (sic., likely the 16/18-gun *Shahbaz Jihad*) to Constantinople,[69] ostensibly in protest over Muhammad Ali's brutal policy on Crete, but according to Cerisy, having already exchanged 65,000 francs into Turkish currency, and later admitting premeditation.[70] There is some confusion about Osman Pasha's subsequent life, which warrants brief attention for its implications for Egypt's involvement in the Crimean War. The most common version found in older Arabic and English sources traces back to the two-volume account of the British physician William Holt Yates, who claims that Osman died in Turkey soon after, supposedly of the plague but with murder suspected.[71] The alternative version is that he enlisted in the Ottoman fleet, commanded a ship of the line during the bombardment of Acre, eventually reaching the rank of Patrona and commanding the Turkish-Egyptian squadron at the Battle of Sinope. This account is primarily asserted by MacDougall but seems to (no direct citation is given) ultimately trace to the contemporary German encyclopaedia Pierer's Universal-Lexikon.[72] Even accepting the latter's biography of the Ottoman commander at Sinope to be entirely correct, however, merely affirms that the Osman Pasha in question served in the Egyptian Navy while leaving most autobiographical details (date of birth, career course, entry into naval service) incompatible with the influential Uthman Nur al-Din (Osman Nourredin).[73] Thus, it seems most likely that this is a case of name-based conflation of two separate individuals.

These back-and-forth desertions and continued mutual influence illustrate the late-Ottoman world in which the mid-century Egyptian Navy must be situated, especially its senior commanders, who seemed to have considered themselves (and been considered by their Turkish counterparts) as being involved in a civil conflict, or at least one within a shared imperial context. Politically, this was formally confirmed with the end of the Syrian conflict in 1841 in the form of a firman of investiture from the Sultan, confirmed by the British, Austrian, Russian and Prussian representatives.[74] In exchange for the right to hereditary primogeniture, Muhammad Ali was forced to accept (among others terms): the reduction of his territory to Egypt and his army to 18,000, compliance with Ottoman laws and international treaties, tribute payments, loss of the right to appoint senior officers, the placement of Egyptian forces at the Porte's disposal and, most relevantly, that any warship constructed for the Egyptian fleet would require the explicit permission of the Porte.[75] This, combined with the return of the defecting Ottoman vessels, essentially precluded the development of any naval fleet above the rank of an Adjacent Force Projection Navy (rank 5 of 9 in Grove's schema), as the court at Constantinople would naturally not allow the growth of a regional threat (nor supply the Syrian timber it now controlled), and now enjoyed the backing of most of the European powers and especially Britain.

The early 1840s were a particularly difficult time for the fleet. Several naval officers took the opportunity of the Turkish fleet's return to desert. 5 ships of the line were disarmed, and 3,000 sailors were released.[76] The decaying fleet was described by the French traveller Gisquet in 1844 as painful to behold. The ships were stripped of guns and rigging and resembled English prison hulks, "sad colossi" "condemned to rot in immobility and silence" for Egypt now needed no more than a dozen steamers.[77] Officers and men were employed in dockyard and construction work, with progress made on replacing ropes with chains and the building of fortifications and docks.[78] Only in 1845 would fighting ships be launched anew in Alexandria, and to nowhere near the same scale as before. With hindsight, then, the history of Muhammad Ali's navy illustrates how, within the diplomatic context of mid-19th-century regional politics, seeking raw naval power proved a red herring – and, ironically, how, despite being less impressive, the Navy under his successors was no less strategically valuable.

Powerless Pioneer: The Egyptian Navy amidst Technological Change

By the 1840s, the global transition to auxiliary steam propulsion was well underway, which heralded the first of a series of naval technological transitions that spelt disaster for navies without a strong industrial base. While the earliest steam warship (the American floating battery *Demologos*) was launched in 1815, the first purpose-built ocean-going steam warships did not appear for another 15 years. Initially, the inherent limitations of the paddlewheel inhibited its effective employment on larger warships. The location of paddles on both sides of a ship's hull severely limited a vessel's effective broadside while introducing a new vulnerability, and when not in use, paddles greatly affected a ship's performance under sail. Most significantly, the overall expense of constructing and operating paddle frigates soon exceeded that of ships of the line, despite their firepower rarely exceeding 30, albeit generally larger guns.[79] Nevertheless, the wind-independent speed advantage of paddle steamers led to the introduction of a slew of paddle warships in the 1830s, with the Egyptian Navy being a notable early adopter.

The frigate *al-Nil* (Nile), launched in London in 1834, may plausibly lay claim to being the first designed for combat.[80] While only carrying around 50% of the armament, equipment and stores of a similarly sized sailing ship, its large displacement still allowed for a powerful armament befitting a combat frigate of 2 Paixhan shell guns and 20 carronades.[81] In service, *al-Nil* was primarily favoured for its speed and was not recorded to have ever received its full artillery complement, being initially employed in the transportation of mail in the Mediterranean. In 1848, it was listed as one of four steamers forming the fleet of the first Egyptian passenger and mail steamship company.[82] Though called back to arms for the Crimean War under the command of the English-trained Abd al-Hamid Afandi (al-Diyar Bakri), her luck held, and she was uninvolved in any major action.[83] The pioneering ship was recorded as late as 1863 as part of a batch of ships sold to the commercial Egyptian Company, by then almost 30 years in service.[84]

Thanks to its evident advantages over the paddle wheel and the availability of existing infrastructure, the naval adoption of the screw propeller had a revolutionary effect on shipbuilding. While the completion of the first purpose-built steam warship in 1843, the frigate *USS Princeton*, did not trigger an immediate response, developments in Europe were ongoing that would lead to a fully-fledged naval arms race across the English Channel. In 1850, the first screw-driven ship of the line, the *Napoleon,* was launched in France, triggering a British response in the initial form of the *Agamemnon*.[85] The decisive advantage offered by speed unconstrained by the wind, with far less compromise, was not lost on contemporaries. The British Fourth Sea Lord, Alexander Milne, wrote in 1850 that 'ships of the line are useless for checking the rapid movements of a steam squadron', and would merely 'be a drag on any operation'. By 1857, the number of converted and purpose-built steam ships of the line reached 28 and 24 in the British and French fleets, respectively, leaving the rest of the world scrambling to respond.[86] While the rapid development of naval technology in the following decade would render these sizable fleets obsolete almost upon arrival, they nevertheless created a dramatic imbalance in the decade of the 1850s, as was shown during the Crimean War.

Despite the continual development of screw technology, the next steamships constructed for the Egyptian fleet remained small paddle steamers that joined the *al-Nil* in its transport role. The *Bulaq* was launched in 1837, followed by a steamer for the Red Sea and two unarmed ships for the Nile. Military steamer construction resumed with the *Barwaz al-Bahr (Pervaz-i Bahri), Asyut*, *Jaylan al-Bahr*, and *al-Abbasiyya*, built in Alexandria from 1845-1850.[87] Little detail is available on these ships, save that they produced 220 horsepower and were equipped with British machinery, with the

Asyut specifically noted as having tubular boilers and a two-cylinder engine from Maudslay, Sons and Field.[88] The next steam frigate to be launched would, however, be a worthy successor in innovation to the *al-Nil*. This was the *al-Sharqiyya* of 1,477 tons burthen (approx. 2,000-2,200 tons displacement), a screw-driven frigate matching any then in service. Built in Alexandria under the direction of the Turkish-born, Sheerness-trained chief constructor Muhammad Raghib al-Islambuli, the ship was launched in 1847 before sailing to London in 1848 to be fitted with engines from Miller and Ravenhill, carrying its designer, along with 21 carpenters sent for training.[89] This in-service date puts *al-Sharqiyya* within a year or two of the earliest Royal Navy screw frigates and, indeed, she compares favourably in speed with the contemporary *Amphion*.[90]

The service history of *al-Sharqiyya* highlights the late Ottoman web in which the Egyptian Navy must be situated. Transferred as a gift to Sultan Abdulmecid in 1850, she was renamed *Muhbir-i Surur* (Informant of Happiness). Though the British naval historian MacDougall mentions the curious detail that she was later reconverted into a pure sailing frigate, this is not corroborated by Turkish historians, who explicitly make mention of both a lifting-screw design (which would render blocking up the screw-hole largely redundant for sailing efficiency) as well as powerplant issues in 1875, which suggests she remained a steam frigate.[91] Though transferred to Ottoman service, *Muhbir-i Surur* served alongside *Egyptian* contingents in the two major wars in which she was engaged. During the Crimean War, she avoided the disaster at Sinope and is instead recorded as part of a flotilla of fast transports (and likely an escort given her size and firepower) alongside several smaller steamers, delivering ammunition from Constantinople to the Caucasian front.[92] By the 1870s, she was undeniably obsolete but remained in good enough condition to be selected as the seagoing training ship for the Ottoman Naval Academy. Though Dal remarks on her unsuitability for this role as an “old and heavy frigate” with limited power and range, it was likely this very size and obsolescence that recommended her for the role, a not uncommon fate for wooden steam frigates in smaller navies.[93] Moved to the Black Sea in 1876 as part of the buildup to the 77-78 Russo-Turkish War, and possibly reclassified as a Corvette, she once more offered auxiliary service to a joint force alongside Egyptian ships and troops.[94] Designated part of the Black Sea Wooden Ship Squadron, she was tasked with the transportation of troops from Trabzon to Varna, where the Egyptian contingent was initially concentrated.[95] Remarkably, she received a refit as late as 1892 and was only laid up in 1894, a testament to the engineering, ship-building prowess and ingenuity of the mid-century Egyptian Navy.[96]



Figure 2: Rudolph Ackermann, William Foster, and Thomas Goldsworthy Dutton, The Egyptian Steam frigate Sharkie 36 guns, coloured lithograph, 1849. National Maritime Museum, Greenwich, London, PAH0946.

A*l-Sharqiyya* would prove to be the last technical triumph of the 19th-century Egyptian Navy. This was not necessarily for lack of trying, and indeed, Khedive Ismail would develop something of a reputation as an easy mark for Western technological snake-oil salesmen.[97] Domestically, the Egyptian government continued to invest in the education of engineers and technicians, who from the 1850s were increasingly appointed to important state posts, presaging the technocratic dominance of experts in the 20th century.[98] And though attempted acquisitions of armoured vessels from foreign shipyards were blocked by the Porte, with the compulsory purchase of the 2,000-4,500 ton ironclads *Asar-i Tevfik* (ordered as *Ibrahimiyye*), *Asar-i Şevket* (ordered as *Kahira*), *Necm-i Şevket* (ordered as *Muzaffer*), *Lütf-ü Celil*, *Hifz-ur Rahman* and *İclaliye* during their construction in foreign yards, this did not preclude the acquisition of unarmoured, but for their time, state-of-the-art warships.[99] The best example of the latter is the *al-Ibrahimiyya/Ibrahim*, a 4,700-ton unarmoured iron frigate built by La Seyne in 1868, which resembled in concept and capability more the large colonial cruisers (role and later designation) of the Royal Navy than the typically smaller, more conservative vessels then preferred for French domestic use.[100] In service, she seems to have fulfilled this "foreign station flagship" role well in the Red Sea, her presence and size/capacity enabling the Egyptian acquisition and occupation of the ports of Suakin and Massawa.[101]

The technological competitiveness of ships like the *al-Ibrahimiyya* neither reflected a pioneering, forward-looking vision as the al-Nil did, nor a triumph of domestic engineering, design, and shipbuilding as with the *al-Sharqiyya*, but rather the competitive nature of the international arms market. The problem was systemic and industrial, brought about by specific changes in naval architecture, and this failure to keep up with global naval engineering should only be understood in a global context, for an inability to construct, let alone design cutting-edge warships afflicted all

non-Western (and many "Western" navies) in the latter half of the 19th century (and all save Japan in the first half of the 20th); a point lost in the standard shallow narratives of political malignment due to the personal biases of the Khedives.[102]

While this shift in international naval construction dynamics can, in broad strokes, be ascribed to rising technical complexity and differences in industrialisation, the specific moment of divergence in shipbuilding was the adoption of iron as the main material in shipbuilding, which necessitated the existence of an interdependent industrial web that in due course would be labelled the military-industrial complex. Wooden ship construction was and remains a complex and labour-intensive process, with ships like the *al-Sharqiyya* representing the culmination of centuries of development in material science, ship design and engineering practice (diagonal framing/seppings system, iron knees, and rounded sterns, etc.).[103] Nor were late 19th-century warships necessarily more expensive than mid-century ones. In purchasing power parity terms, the cost per ton of an early pre-dreadnought only doubled that of a mid-century wooden vessel, despite the technological gap involved. For industrial states, this in fact represented a significant decrease in relative cost per GDP, which offset the economic burden caused by the increase in overall ship displacement.[104] On the opposite side of the equation, the industrial and corresponding terms of trade differentials that created the "West and the Rest" great divergence dynamic were already strongly entrenched for decades by 1860, even in the most conservative accounts, yet despite the massive gulf in social-economic development, states like the Ottoman Empire and Egypt could still design and build cutting-edge wooden vessels.[105]

The key differentiating factor was thus in the shifting structure of costs. Even in its final hybrid-propulsion form, the upstream industrial requirements of wooden warships remained limited and compartmentalised to several areas such as structural reinforcements, masts, engines, and artillery. Wood, while regionally constrained, was ultimately naturally occurring and could be employed by conscripted labourers and artisans.[106] Thus, a state without an industrial economy could still construct state-of-the-art wooden warships by limiting imports to specific areas and leveraging its cheap labour to achieve an advantage in purchasing power parity.[107] The same could not be said for the construction of iron-hulled vessels, let alone ironclads. Non-Western states without an established industrial base in the latter half of the twentieth century were essentially faced with two options for short and medium-term naval acquisition, both of which exposed them to market exchange rates that their less productive economies and fragile fiscal structures could ill-afford. The standard and more conservative approach was industrial surrender: the purchase of warships from foreign yards, or, if naval combat was not a primary concern, second-hand vessels from other navies. From a purely military perspective, this allowed non-industrial states to tap into the established economies of scale and accumulated industrial know-how of leading industrial nations at a manageable premium in cost. The disadvantages of this approach were long-term: geopolitical risk from dependency, vulnerability to delivery complications, and the further atrophy of local military shipbuilding and related industries.[108]

The alternative was to attempt to keep pace, but in this, the growing upstream industrial requirements of iron-hulled shipbuilding rendered the narrow dockyard/arsenal programs of the early 19th century insufficient. Instead, a wider program of military adjacent infrastructural industrialisation from coal mines to iron works to transport infrastructure was now a prerequisite, in addition to specialised arms factories for engines, guns, etc, the capital machinery and knowledge for which had to be imported at market rates.[109] Such a program was by its nature complex, protracted and uncertain, and highly sensitive to political interference both domestically and abroad due to the need for continual investment. For states without a civil industrial base, this financial

burden and corresponding risk would essentially have to be borne by the treasury. As such, only three states would attempt such programs in the late-19th century to mixed results – the Ottoman Empire, the Qing Empire and Imperial Japan. In the first two instances, the naval construction programs, while delivering some results, never received the sustained investment to create the necessary momentum to achieve sustainable industrial autonomy, and are therefore generally evaluated as expensive failures.[110] In the latter case, the case can be made that the national mobilisation required to sustain the naval and military build-up of the 1880s was a major factor behind the hawkish policies that led to the continual imperial expansion, which only terminated in 1945.[111]

Perhaps fortunately for Egypt, wider Ottoman politics in the 1860s precluded any such attempt at large-scale naval modernisation, though the issues inherent in the introduction of specific naval industry within a generally unindustrialized state and society nevertheless manifested in microcosm. The engines of the steamship fleet, for instance, required continual maintenance from English engineers.[112] The most illustrative example, however, was the Egyptian torpedo program, which ran for a decade from 1871 to 1882. Interest in this obscure project has largely been limited to laudatory American historians owing to its US Civil War connections, with little attention garnered outside of this historiographical niche.

For clarity, the following discussion will use the modern definitions when simply referring to torpedoes and mines, and avoid the term submarine, though note that in the mid-late 19th century, "torpedo" was often used generically for all underwater explosives before 1870 (as in "damn the torpedoes") and submarine generally meant all self-propelled underwater vessels and ordnance.[113]

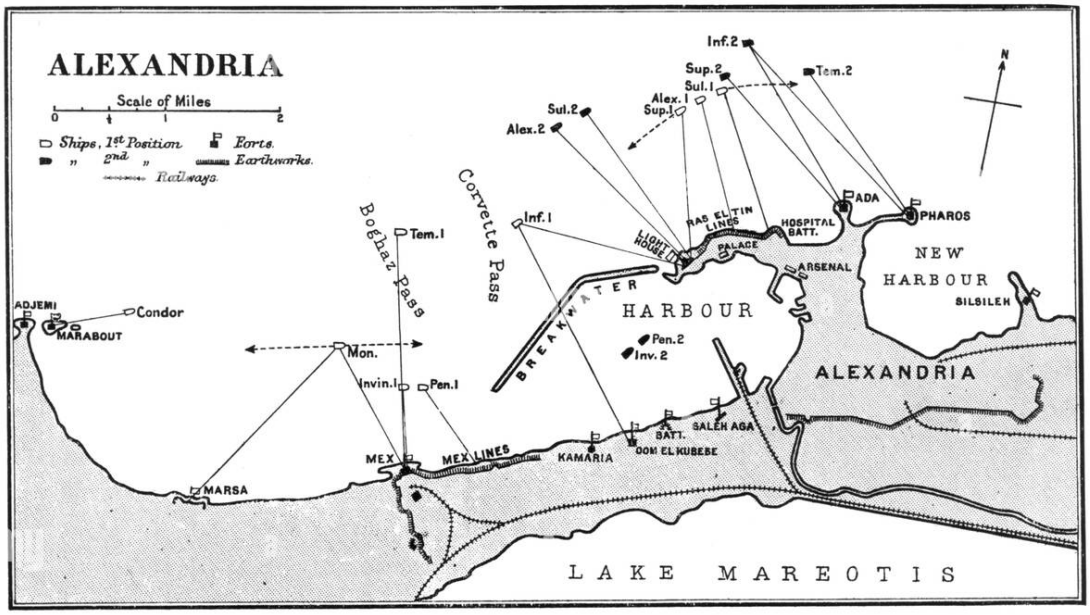
The key figure in Egypt's torpedo program was John Louis Lay, one of a select few pioneers of torpedo technology of the 1860s and 70s whose work may be considered foundational to the weapon's eventual maturity towards the turn of the century. Before he arrived in Egypt with the American military mission in 1871, Lay already possessed a reputation courtesy of his co-invention of a spar torpedo (essentially a bomb fitted to a long pole), which sank the Confederate ironclad ram *Albermarle.*[114] Together with fellow US Colonel William Ward and under the direction of Ismail's American chief of staff Charles Pomeroy Stone, he first surveyed the Egyptian coast in search of suitable locations for installation, then in 1873 began planning for a naval school at Rosetta to train 50 officers in mining and torpedo operations.[115] For the equipment, Lay was given orders for mechanical mines and automatic suspension mines, which were to be arranged in a wired net to be simultaneously detonated either through contact by a target ship or command from a shore-based operator. More radically, a self-propelled torpedo that could be controlled through electric cables on shore, with a claimed range of two miles at eight knots was also purchased and delivered to Egypt.[116] The cost of these systems was considerable – for the purchase of a batch of 5 self-propelled torpedoes, three automatic mines, and the exclusive right to manufacture, Lay (who had since departed Egypt) was given £23,000 in 1874, which, for context, constituted slightly more than half the cost of a thousand-ton Royal Navy sloop at the time.[117]

While Cox, despite his empirical rigour, seems to accept at face value the platitudes and claims made by official statements and self-motivated participants regarding the capabilities of Lay's weapons, describing the systems as formidable defensive naval weapons thwarted merely by the "cleverness of the English", the tests of war and peace in the following decade conclusively demonstrated the contrary.[118] The first use of the Lay Torpedo took place halfway across the globe in 1879, when the Peruvian ironclad Huascar attempted to attack Chilean transports and warships anchored in Antofagasta with the projectile. Once launched, the torpedo proved wholly

uncontrollable and instead double backed towards the Huascar, with tragedy supposedly only averted after the young lieutenant Fermín Diez Canseco dove into the water to physically divert the device. Frustrated, the Peruvians refused to pay for the weapons, with the commanding admiral Grau dramatically burying the remaining torpedoes at a cemetery as a show of disdain.[119] While the Peruvians would eventually achieve some success with conventional mining, Lay torpedoes were largely absent from these scores. Just as illustrative of the weapons' limitations was their dismissal by the world's major navies despite their presence at multiple demonstrations. The opinion of the Anglo-Ottoman Admiral Hobart Pasha in 1882 is representative – following a series of trials at the Bosphorus, he cast doubt on the veracity of Lay's wild claims, observing that the weapon was inadequate for use at sea, with its only advantage over the Whitehead series torpedo being in rivers and estuaries with strong currents.[120] This, combined with the weapons' great complexity and questionable build quality, must be factored into analysis of the failure of Lay's defensive systems during the Bombardment of Alexandria.

The best-known story regarding the Egyptian failure to employ torpedoes in their 1882 defence is recounted by Kusel Bey (Baron de Kusel), the English acting-director of Egyptian Customs, in his autobiography *An Englishman's Recollections of Egypt*. Kusel recounts how, in the tense lead-up to the British invasion, he received numerous applications from Urabi and his war office subordinates regarding the arrival of a package addressed either to General Stone Pasha or the minister of war. Suspicious, Kusel elected to covertly investigate whether the package existed and what it might contain alongside a European inspector, eventually discovering a case containing a torpedo hidden beneath other packages. After instructing the inspector to sabotage the weapon, Kusel consulted with the British comptroller general Colvin, who ordered him to deliver the weapon to Admiral Seymour aboard the *Invincible*, thus removing any possibility of the weapon's employment.[121]

Despite dominating the literature, Kusel's intervention alone should not have been sufficient to neutralise the Egyptian underwater defences. A sizable stock of mines was found after the bombardment at Fort Mex, one of the heavily engaged positions during the battle. These included 87 500-pound gun-cotton mines, 87 250-pound gun-cotton mines, and 500 100-pound electro-contact mines, with the only lacking supplies being cable, of which only two knots could be found following the battle, according to Goodrich, an American observer with the Office of Naval Intelligence.[122] None of the mines, however, were laid before the engagement began, and while Goodrich's report attributes the cause of the failure to deploy the mines to a "lack of experience and competent *personnel*" (italics in original), British intelligence suggests that surprise was just as significant of an issue; as in addition to the mines, boats were supposedly being readied to be sunk with rocks to block the "Corvette Pass", with



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Figure 3: Map Showing the Bombardment of Alexandria, Royal Armouries Museum (reproduced from H. L. Wilson, Ironclads in Action (1896), p. 340).

a gunboat being directed to patrol for wires at night in response, but neither plan was carried out in time.[123] The latter accusation was, however, denied by Egyptian officials as erroneous and technically impossible, and in any case easily within the British ability to prevent.[124]

With hindsight, the Egyptian torpedo program encapsulated the major pitfalls faced by military industrialisation programs of non-Western States in the late 19th century. The most evident was the political risk inherent in reliance on foreign nationals – if Kusel's actions can be put down to divided loyalties, then the departure of Lay and much of the American military mission, which seemingly deprived the project of its remaining momentum, is representative of the sustainment risk inherent to foreign-led programs lacking a sufficient base of organic local support. This pattern was not unique to Egypt, being repeated in the mass acquisition of flat-iron/Rendell gunboats by the Qing navy,[125] and the self-defeating extreme designs of vessels such as the IJN's *Matsushima* class protected cruiser.[126] The Egyptian Navy would only return to the international technological forefront more than half a century later, when in 1967 it inaugurated the end of the 4-century domination of naval artillery with the first successful surface-to-surface anti-ship missiles engagement, sinking the Israeli Destroyer *Eilat* with Soviet *P-15 Termit* missiles fired from *Komar*-class missile boats, a topic better left to students of the Cold War.[127]

If the Egyptian Navy in the latter 19th century fell behind technologically, its continual technical investment and built-up technical expertise were not in vain. As the following chapters demonstrate, the force continued to operate effectively across the Black Sea, the Mediterranean, the Red Sea and the Indian Ocean, projecting seapower in support of Egyptian strategic aims even if, in raw naval power terms, it was relegated to strictly second-tier status.

"Sons of Mamelukes": Naval Education and Recruitment

Part of the European dismissal of Egyptian naval capabilities is a belief that the combination of compulsory recruitment, limited training and poor service conditions could not but result in an uninspired performance by the fleet's sailors at the best of times, echoing wider orientalist notions of despotism and its deleterious effect on military effectiveness.[128] While little information is directly available on the recruitment of rank-and-file sailors, the sources indicate that the conscription pipeline was shared with that of the army.[129] Initially, this was a haphazard and violent process in which officers descended among villages and simply seized men regardless of their age, marital status or number of brothers, tying them together by the neck in groups of 6 to 8.[130] Gradually, the system stabilized, with the responsibility of supplying conscripts delegated to local village sheikhs. This granted them considerable power, the abuse of which, whether through favouritism or corruption, became a common complaint in Egyptian village life.[131] If more orderly, recruitment remained a coercive and involuntary process throughout the 19th century, with desertion the most common form of resistance. With naval service supposedly even more detested than that on land, extensive measures were required to prevent escape,[132] including the tattooing of sailors on their arms and legs with tattoos of a ship, a fish or an anchor (as was done to convicts),[133] a compulsory system of passports/certificates for internal movement, the tasking of dedicated units such as sappers and Bedouin irregulars to police recruits and hunt for deserters,[134] and the construction of clearly demarcated and isolated barracks and camps from which soldiers could not leave without a stamped certificate.[135] If caught, deserters faced whipping, hard labour, loss of pay, imprisonment, and confiscation of property.[136] This military system of control was what Mitchell famously described in poststructuralist terms as the introduction of colonial and capitalist modes of discipline and power to Egypt.[137]

Not all of the rank-and-file sailors were recruited from the villages of Egypt. Greeks and other Europeans also served in the ranks, and the French takeover of Algeria saw an influx of 1,200 to 1,500 personnel from the former Dey's fleet.[138] As for officer recruitment, this was largely bifurcated between Egyptian-born and foreign-origin personnel on the one hand, and between technical and command roles on the other. In the former category, the early career of Ahmad Urabi illustrates the basic process of selection. Conscripted at the age of 14 around the year 1854, he was explicitly targeted for officer recruitment as a "son of the village Sheykhs", having already undergone some education at the traditional Collegiate-Mosque of al-Azhar in Cairo. He was then offered the junior administrative position of a clerk after an exam, a dead-end position that the ambitious young officer turned down.[139] In the latter privileged category, though European hires are disproportionately represented in the literature, most recruits came from within the Ottoman World. The English-trained administrator Yusuf Hekekyan, himself of Armenian origin, describes the officers of the Egyptian fleet in 1834 as "Circassians, Abyssinians, and stolen [enslaved] Colmen [likely an anachronistic reference to Cumans] and Tartars."[140] Somewhat ironically, Greeks played an important part in the development of the Navy, with the Zizinia merchant family providing the Wali's first frigates, whose captains in turn often hailed from Greek islands, some of whom later converted to Islam.[141] Even after the war in Greece, around 60 Greeks were still recorded in Egyptian service in the early 1830s.[142]

The specific label of "foreign", however, must be heavily caveated, as this classification seems to have generally been applied to all Turkish-speaking members of the elite regardless of geographical origin. The divide was largely socio-cultural and linguistic – a combination of internalised differences in dress (e.g. the Fez and other European-style clothing), etiquette (bland and "prepossessing"), transportation (e.g. the use of horses and carriages), and the use of Turkish

instead of Arabic.[143] Command of Turkish thus did not automatically confer elite status – instead, the "children of Arabs" (evlad-i Arab) remained in subordinate positions within the hierarchy regardless of merit. In contrast, those labelled "mamluks", by this era a well-entrenched group in Egypt, were often by the mid-19th century neither Mamluks in the strict sense of possessing military-slave status nor necessarily primarily Turkish speakers but were nonetheless still set aside by their ancestry and membership in the elite household patronage system.[144]

That many of the sailors in the fleet served involuntarily should not imply that training was always lacking or poor. In fact, considerable resources were dedicated to training activities, particularly after the defeat at Navarino. Naval training was concentrated in Alexandria, either in the district of Ra's at-Tin, or from 1830 onwards, aboard larger warships. At the bottom of the hierarchy, a school was set up for the large-scale training of 12,000 sailors in ship handling and naval gunnery.[145] Officers received more extensive training. Ensigns and petty officers were taught navigation and seamanship in dedicated schools aboard ship, while those in the privileged class destined for command were to be formally educated at the *Madrasat an-Nawatiyah (Naval School)* in arithmetic, geometry, geography and navigation under both European (often French and Italian) and local teachers, before their despatch aboard ships to obtain practical experience.[146] At the top of the hierarchy, those groomed for senior command were often sent abroad on educational missions in Europe to receive naval training, the first being Uthman Nur al-Din (Osman Nourredin), who introduced French naval codes and became the commanding admiral of the navy, in marked contrast with the experience of Egyptians educated abroad, whose prospects were usually limited to technical administration.[147] The sons of Muhammad Ali, meanwhile, received personal specialist advice and instruction from foreign experts on naval warfare – the French captain Jean-Marie Letellier was hired to advise Ibrahim, while Sa'id, who was enrolled in the navy, was placed under the general tutelage of Jean-Victor Besson, replaced after his death by Houssart Bey, another former French naval officer.[148]

A rare glimpse of daily life for the rank and file in the Egyptian Navy during the reign of Abbas was captured by an anonymous Russian visitor in 1851, who spent two weeks in Alexandria and published his observations in the official journal *Naval Collection (Morskoi sbornik)*. The identity of the writer can be inferred to be likely a junior naval officer aboard a visiting warship, from the lack of any official tasking/reception mentioned here or in other sources, his familiarity with Russian naval practice, the mention of being anchored for the duration, and the professional nature of the publication. At the highest level, the visitor describes the fleet as one of the "Turkish Empire", confirming the tripartite divide between the Ottoman (non-native) origin officers, Europeans and natives of Egypt. Europeans were forbidden from the ranks and only recruited to fill specific technical roles, while Egyptians could not rise above midshipmen.[149]

Overall, the writer carried a mixed impression of the fleet's personnel, with enlisted men comparing favourably to their officers. The lower ranks are described as cheerful and efficient, able to complete basic exercises such as the readying and operation of the guns and the raising of sails capably. They were well dressed in modern Ottoman attire: white canvas trousers, a white single-breasted jacket with a black anchor painted on the left sleeve, shoes, and a red fez with a blue tassel. The days of the sailors were mostly spent in training or work in the harbour, as well as much busy work cleaning and polishing their vessels, whose cabins and decks were spotlessly clean. At night, they were housed aboard their warships for lack of accommodation on land, and approximately half the crews with relatives in the city were allowed to spend nights ashore.[150]

The “Turkish” officers were judged more harshly, primarily for their lack of technical competency, the observer describing an anecdote in which an officer instructing in navigation believed the maps the Russians used were inaccurate as they showed compass declination differently in different locations, and was incredulous when the reason was explained to him. Junior officers were preferably recruited from the children of naval personnel, but sons of farmers filled the ranks. Apparently, their training curricula had shrunk from a broad program including foreign languages to just reading and writing in Arabic and Turkish and superficial navigation, conducted aboard ship as shore-based facilities were no longer available. As graduate midshipmen, they had little hope of promotion and were most often observed bringing their superiors coffee and pipes.[151]

The largest issue observed with the fleet was its staticity, with most vessels being permanently anchored in the roadsteads off Alexandria. New ships were not being built, and the vessels in service were described as rotting, being repaired slowly and insecurely; this had reached such an extent that most ships had not been at sea for 10 years and no major vessels for the previous 2 years, causing the writer to cast doubt on the actual capabilities of its personnel.[152]

The reign of Abbas is often considered to be a nadir in Egyptian naval development, and this view seems to hold true at least in the sphere of naval education. To reduce costs, the naval school in Alexandria was shut down in 1849, along with the infantry and cavalry schools, with higher military education consolidated into a single school with the name of “chosen” (al-Mafruza). This centralised institution was a primary, preparatory and military school rolled into one, and included a distinct engineering aspect, but naval education seems to have been lacking.[153] Described by Baron von Malortie as a nursery for army officers, and implied by Hekekyan to contain boys whom Abbas found attractive, the school at least seems to have had some success educating the pick of the students from the shuttered institutions and grew to more than 1,000 students in size, Sarhank even describing it as a great success.[154] For his part, Abbas seems to have been aware of the negative international perception caused by this “regression” (and the laying off of highly paid foreigners), writing to French officials that he was merely seeking to focus on foundational primary education and redressing abuses, with developed industry and advanced schools of little use in a country before basic needs were achieved.[155]

The subsequent reign of Said saw a resurrection of Egyptian naval education. In 1860, the Alexandrian naval school was reopened under an Italian Captain Federico, who held the post from January 1860 to August 1863.[156] While the grander hopes caused by Said’s extensive connection with the Navy did not come to pass, funding was nevertheless made available to continue the transition to steamships, and naval students were placed aboard the vessels of Egypt’s growing merchant marine in a bid to further their navigational skills.[157]

Under Ismail, the educational apparatus was expanded and formalised. A second school was opened at Alexandria, likely for shipbuilding, while the naval curriculum was concretely recorded as three years and covered the basic subjects of European officer training. These can be divided into three main categories: seamanship, weapons handling, and theoretical knowledge necessary for command. In the first category were chart and globe-reading, navigation, rowing, sailing, rigging, naval manoeuvres, as well as swimming; in the second, the use of swords, artillery, as well as mines; and in the final category, the English and Turkish languages, military laws, geography, physics, trigonometry, and the solution of naval problems. Many of the subjects, such as the use of artillery, were covered across all three years, likely indicating both staged progression and a relatively high course intensity. One prominent graduate of the schools in this period was Ismail Sarhank Pasha, who completed his studies under the supervision of the equally storied British

school director McKillop Bey. Himself the son of a naval officer, Sarhank would later rise to direct military schools and enjoy a successful naval career, but remains best known for his historical work, *True Narrations of Maritime Nations*.[158]

In the historiographically contested area of the promotion of "native" officers versus "mamluks", the transition from Abbas to Said is generally viewed progressively, while the policies of Ismail are considered a regression. While not inaccurate, this narrative fails to capture the extent to which the Turkish and Circassian elites remained preferentially treated, and the continued limited promotion of Egyptian officers as a political device or to resolve shortfalls in personnel. The primary reason for the "regression" in Abbas's reign was likely the reduction in the size of the armed forces, with the Army cut to 18,000 men, in which preservation of the senior officer corps took precedence.[159] The reign of Said, of whom it was said "the army was his hobby", saw more extensive opportunities for promotion opened to native Egyptian once more, including the expansion of conscription to the previously exempt children of village chiefs, as seen in the aforementioned recruitment of Urabi into a junior officer rank.[160]

Under Ismail, these Egyptian officers once more saw their prospects frozen, though due to the shortage of non-native origin officers, a select few were able to reach as high as the rank of lieutenant-colonel (qaimmaqam).[161] The formerly conscripted sons of village chiefs were now allowed to buy themselves out of military service.[162] It should be noted, however, that in his wider administration, Ismail furthered the inclusion of Egyptians in the bureaucracy, governorships, cabinet positions, as well as the newly created chamber of deputies, and their continued barring from military high command is probably a better reflection of the uniqueness of wider late Ottoman military culture and its criticality to the Khedives' power base than any generalised hostility.[163]

Broader conscription practices, at least, seemed to have become better ordered and generally milder over time. From the reign of Abbas onwards, all able-bodied males were eligible for the draft, with conscripts selected through the casting of lots with quotas assigned per village. Bedouins, students and from 1863 economic hardship cases were exempted, though notably unlike in the Ottoman Empire, non-Muslims like Copts and Jews were formally conscripted from 1858. The rich could still avoid service by paying a sum of money, or by providing a substitute in the form of a black slave. The Russian observer discussed above was told by officials that the term of naval service was ten years, but that in practice, discharge only came due to old age or injury.[164] While the term of conscript service was ostensibly reduced by Said to a year and a half, it is unclear whether this applied to naval service.[165]

Just as significant for the build-up and sustainment of the fleet as its sailors were the large cohorts of construction workers, carpenters, craftsmen and engineers who laboured on its vessels, dockyards, machinery and wider supporting infrastructure. The developmental history of Egyptian naval architecture in the 19th century mirrored the fleet it supported. In the first phase, ready know-how and expertise were primarily imported from abroad through the hiring of European specialists and the dispatch of students overseas. Italy played an especially prominent role in this early phase: the first arsenal built at Alexandria in 1826 was primarily staffed by Genoans, with administrative and technical roles filled by the Vaccarini, Avoscani, and Demarchi families, and Italian was observed to be a dominant language in 1830.[166] The peninsula was also a popular destination for early Egyptian students sent abroad to acquire qualifications in smelting, naval architecture, military arts, and printing, with the latter facilitating the translation of books in the arts and sciences into Arabic and Ottoman Turkish.[167] Nevertheless, the international context of growing Anglo-French naval domination of the mid-19th century increasingly reflected on Egypt as

well, and subsequent returning students and specialists more often hailed from the ports of England and France. Toulon was particularly favoured, notably represented by the French shipbuilder Louis-Charles Lefébure de Cerisy, whose 7-year tenure heading the Alexandria Arsenal from 1827-35 represented its most successful.[168] As for the recruitment of labourers and artisans for work at the dockyards, they seem to have either been conscripted in the same manner as military conscripts, while convicts made up the labour at the "Liman", as the prison arsenal was known (and which became synonymous with prison in Egypt).[169]

The period of cuts from 1840 onwards, followed by the fiscally conservative regime of Abbas, seems to have put a dampener on naval construction, with many foreign workers returning to Europe. Naval personnel were dispatched instead to work on infrastructure construction projects, such as railway bridges for the new railway connecting Alexandria to Cairo.[170] Only in the reign of Ismail did there appear to be something of a renaissance – students were once more dispatched to Europe to be educated in naval architecture, and additional naval workshops, docks, and corresponding training facilities were constructed.[171] This lasted until the economic collapse of the late 1870s, when, under pressure from European bondholders, Egyptian spending was dramatically curtailed, and the government schools shuttered,[172] with the Navy itself being abolished in 1884 under British administration.[173]

In the final analysis, Egyptian naval education and recruitment policies in the late 19th century were a qualified success, keeping the fleet in a reasonable state of operational readiness and competence, within a broader statewide pattern of societal technical modernisation. Nevertheless, the organisational shortfalls of ethnic inequality, unpopular conscription practices, and reliance on Western expertise remained unresolved, though unlike in the Army, they remained under the surface until the disbandment of the Navy itself.

But To Do and Die: the Egyptian Navy in the Crimean War

Though the reign of Abbas I is typically associated with the continued diminution of the Egyptian Navy following the second Syrian War, it also saw the fleet's last major foreign deployments, as well as its last surface engagement. Through a detailed examination of the fleet's activities during and immediately preceding the Crimean War, this chapter questions whether this narrative of collapse is overstated, as well as how the wider course of late Ottoman politics remained a key determining factor in Egyptian naval history.

An underlying assumption often made in older and popular English histories of the Crimean War is that Egyptian participation was a forgone result of Ottoman involvement, in contrast with the complex diplomatic manoeuvres associated with the Piedmontese intervention (and Austrian neutrality). This is more a result of neglect than misunderstanding, given the generally peripheral role played by the Ottoman Empire in Western accounts, which tend to treat Imperial politics within the broader background of the Eastern Question, and their tendency to focus on the land operations on the Crimean Peninsula at the expense of other theatres and naval engagements.[174]

The basis of Egyptian participation in the war was the 1841 firman, under which Cairo was obligated to provide 10,000 troops for war should the Porte require them. Nevertheless, histories of the Crimean War from a Turkish perspective are under no illusion that the provision of troops from the autonomous provinces was a given; instead, the despatch and upkeep of the Egyptian contingent constituted a continuous negotiation where concessions were expected and granted from the centre to the periphery. The cost of maintaining the Egyptian armies, for instance, was paid from Ottoman coffers, while the excess maintenance costs of the approximately 5,000 men sent from Egypt

beyond the base ten-thousand-man requirement were deducted from the tribute owed to Constantinople.[175]

The Navy occupied a position of unique political sensitivity in this period of Egyptian history. This was primarily due to its leader, Said Pasha, the fourth son of Muhammad Ali and uncle of Abbas I, whose rivalry with his nephew was well known and who was (correctly) perceived as a potential successor. While Said managed to avoid exile and ostensibly remained head of the navy, a range of alleged plots nonetheless plagued relations between the two men. One supposedly involved Said's collection of a force of Bedouins in the Nile Delta for nefarious purposes, which prompted the forceful migration of several tribes to the lower country; Said himself insisted that the clearance was merely for the benefit of Abbas' son Ibrahim Ilhami, contrasting the tameness of his opposition with the European revolutionaries of 1848.[176]

Another plot, uncovered in May 1852, was, however, much better substantiated in Egyptian archival records and Western diplomatic correspondence. This involved the secret assembly and training of upwards of several hundred men on one of Said's estates near Alexandria, comprising naval troops, members of opposition households and their auxiliaries, and Bedouin tribesmen. Five naval officers acted as military instructors. After rumours of unusual activity were reported to the governor, a contingent of sixty men in uniform, with six fully equipped field guns, sixty-three rifles, and a considerable amount of ammunition was arrested. All arrested were eventually sentenced to fifteen years of hard labour in the mines of Jabal Fezzan in Sudan, after an initial lighter sentence was overturned. Despite this, Said was left in command of the navy, though he was forbidden from taking an intended yacht trip to Europe.[177]

An additional point of sensitivity for the Navy arose from its role in countering an alleged plot originating in Constantinople in 1851. This incident, recorded in American consular dispatches, involved the transit of a 1,200-strong Albanian regiment from the Levant, ostensibly as the vanguard of 8 to 9 others, all heading to the Hedjaz to suppress Wahhabi activity. On arrival in Alexandria, however, the Ottoman troops refused to deposit their arms, and to prevent a forced landing, two ships of the line were deployed, training their broadsides on their transports. According to McCauley, the next day an order was received to allow the troops to land, but once they were ashore, the men were made to set up camp outside the walls on the pretext that no barracks were available, whereupon they were surprised by two veteran Egyptian regiments who forced the issue, leaving the Ottoman troops no choice but to turn over their weapons and march overland unarmed.[178]

That the incident occurred amid a backdrop of wider contestation over the extent and limits of regional autonomy fuelled further suspicion. In particular, the question of whether Ottoman approval was required for the railway line from Alexandria to Cairo had yet to be resolved, a question whose answer would only come in the months following British naval pressure.[179] Cairo's response to the perceived Ottoman threat was thus military: 10,000 additional men were sent to Alexandria to monitor any further landings, and the recently expanded defences were brought up to a high state of readiness.[180]

It is therefore in this politically charged context that the Porte's preparations for war with Russia and the calling to arms of its vassals must be situated. While the previous nationalist narratives of "Turkish" passive victimhood have been historiographically discarded, it remains the consensus that the considerable Russian advantage on the tactical, operational, and strategic levels left the Ottoman government in a reactive position bilaterally; a situation made worse by its widespread failure to prepare contingency plans on both the diplomatic and military levels.[181] Thus, following

the terminal rise in tensions following the Russian ultimatum on 5 May 1853, which went beyond the ongoing dispute over the holy places in Jerusalem to demand the acceptance of Russian protection of all Orthodox subjects of the Ottoman Empire, Ottoman mobilisation was limited in scope, which made the orders sent to the autonomous dominions of Egypt and Tunisia to dispatch troops to strengthen the forces on the Danube front all the more important.[182]

From Abbas's perspective, the Ottoman request presented an opportunity on several levels. Diplomatically, it served to justify the continual expansion of the Egyptian Army past the 18,000 men limit imposed in the 1841 Firman; fiscally, the burden to maintain the troops dispatched could now be offset by a reduction in the tribute owed to the Sultan; while politically, the dispatch of most of the navy to the Black Sea must have constituted a reduction in the opposition posed from that quarter.[183] Thus, after brief negotiations, a sizable force of around 20,000 men in 6 infantry regiments, 1 cavalry regiment and 12 batteries was sent to Varna in August 1853 alongside and aboard a squadron of 12 warships. This was followed by a second division in 1854 of 3 infantry, 1 cavalry, and 1 artillery regiments, which was initially diverted to suppress Greek rebels in Epirus, Thessaly, and Macedonia, and a third in April 1855, with approximately 12-13,000 men, which successfully defended Eupatoria against a Russian counterattack alongside the second contingent.[184] The back-and-forth transport requirements generated by this multi-wave movement likely explain some of the inconsistencies regarding the identity of the Egyptian vessels dispatched, with replacement vessels possibly recorded in some sources in place of the originals. The fact that a large naval force could be made ready in short order, meanwhile, contradicts some of the more severe claims regarding the deliberate neglect of the Navy by Abbas.[185]

Even with Egyptian reinforcements, the naval balance of power in the Black Sea theatre heavily favoured the Russians, who possessed 17 ships of the line and 7 frigates in the region compared with an Ottoman total of 14 Imperial and 3 Egyptian ships of the line and 3 Imperial and 4 Egyptian frigates, the Ottoman strength being further diminished by low readiness, with 6 ships of the line and a frigate under repair or construction. Russian naval dominance was thus assured even before qualitative differences in steam adoption, artillery, training and command and control are considered. Within this context, the initial Ottoman naval deployment in late 1853 seems to reflect a political-military balance of risk mitigation on the one hand, and the need to maintain a meaningful presence to cover key ports and coastal lines of communication, on the other, rather than necessarily a “lack of a campaign strategy plan” as described by Turkish historians like Mercan.[186]

The Egyptian Squadron in the Crimean War[187]

Type	Name	Guns	Complement
Ship of the Line	Miftah al-Jihad	100	1,040
Ship of the Line	Halab	100	1,040
Ship of the Line	al-Fayyum/Bani Suwayf?	100	1,040
Frigate	Rashid	60	631
Frigate	Shir al-Jihad (Lion/Leone)	60	631
Frigate	Dumyat (Damietta)	60	631
Frigate	al-Buhayra (Egyptienne)	60	631

Type	Name	Guns	Complement
Corvette	Jinah al-Bahr	24	213
Corvette	Jihad Baykar	24	213
Steamer	al-Nil (Nile)	30	371
Steamer	Barwaz al-Bahr (Pervaz i-Bahri)	12	179
Schooner	al-Saiqa	12	179

At the onset of hostilities, the Porte deployed four squadrons into the Black Sea: one under the fleet commander Bahriye Feriki (Vice-Admiral) Ahmed Pasha, which seemed to have stayed close to Constantinople, a second under Egyptian Mirliva (Rear Admiral) Hassan Pasha, covering the transport routes to Varna, a third under Ferik İngiliz Mustafa Pasha patrolled the Georgian and Circassian coast, and finally a fourth group under Patrona (Vice-Admiral) Osman Pasha to cruise the shores of northwestern Anatolia. Critically, none of the forward-deployed formations contained ships of the line, with the third Eastern Group consisting entirely of paddle steamers with an advantage in mobility and limited firepower. While capital ships were prepared for reinforcement, the order to sail was cancelled, ostensibly on the advice of the British Ambassadors, though given that Anglo-French opinion was generally opposed to the dispatch of any warships into potential battle without Allied support, the disposition of the Ottoman warships in this first phase of war was more likely the result of internal balancing between the risk of exposure to superior forces and the need to maintain lines of communication, rather than foreign interference and/or conspiracy.[188]

The first engagement involving an Egyptian warship in the Crimean War took place on the 17th of November (Gregorian) off the coast of Ereğli, roughly a third of the distance between Constantinople and Sinope. The ship in question was the Egyptian paddle steamer *Barwaz al-Bahr*, better known in Turkish and Russian sources as the *Pervaz i-Bahri*, assigned as a scouting auxiliary to Osman Pasha's Sinope squadron. The *Barwaz al-Bahr* seems to have just finished coaling at Ereğli when smoke from her funnels was spotted by the Russian paddle frigate *Vladimir*, likely on a raiding mission and flying the flag of Vice Admiral Vladimir Alexeyevich Kornilov, chief of staff of the Black Sea Fleet.[189]

In most accounts of the battle, the two ships are described as being roughly evenly matched, but this is almost certainly deceptive. Launched and built in Alexandria from 1844 to 1845, the *Barwaz al-Bahr* was equipped with 10 to 12 "large calibre" guns and fitted with British machinery of 220 horsepower (likely nhp). The Blackwall-built *Vladimir*, meanwhile, was launched three years later in 1848, mounting 11 guns, including 2 10-inch and 3 68-pdr shell guns, and installed with a 400 nhp/1,200 ihp powerplant built by Rennie in London. Her displacement was recorded at 1,731 tons at 200 feet in length. Thus, despite mounting a similar number of guns, the *Vladimir* was far closer in size to the *al-Nil (Nile)* than the *Barwaz al-Bahr*, whose displacement is not concretely recorded, but was likely far smaller given she carried less than half the complement, with the implication that the smaller Egyptian hull could not have matched the weight of the *Vladimir*'s very large-calibre guns. The Russian ship enjoyed a further advantage courtesy of her more powerful machinery, which could propel her to an impressive 11 knots.[190] Given these differences, the oft-cited lack of end-on firepower of the *Barwaz al-Bahr's* broadside layout appears to be more of an additional disadvantage rather than a decisive factor.

The fact that the *Vladimir* was superior and at least as fast as almost every Turkish steamer then in service likely informed Vice-Admiral Kornilov's decision to order her captain, G.I. Butakov, to

give chase upon sighting smoke on the horizon at dawn. The battle occurred on a calm sea, with both ships having full use of sail and steam. The Russian ship benefited from her smokeless anthracite fuel, which made her appear to be a sailing merchant ship, and was therefore able to close the distance without raising alarm. After a four-hour pursuit, the engagement began between 9 and 10 am when the *Barwaz al-Bahr* was overhauled, the *Vladimir* raising the Russian ensign and the *Barwaz al-Bahr* doing the same in response. Taking advantage of his ship's mobility and his opponent's lack of rear-facing artillery, Butakov repeatedly raked the *Barwaz al-Bahr's* stern with the *Vladimir's* five-gun broadside, to which little effective reply could be given.

The battle was concluded after *Vladimir* closed to 100 yards and unleashed volleys of grapeshot. With 22 dead and 18 wounded, including her captain, Saghqul (Commander) Salih (given in Russian sources as Said Pasha, an honorific unlikely to be possessed by the commander of a small steamship), her steering disabled, and her compasses destroyed, the *Barwaz al-Bahr* lowered her flag at approximately 1 pm. The Russian boarding party was supposedly surprised to find the surviving crew smoking pipes among the corpses on deck, with a look of mortal fear and submission to fate. Russian losses were only two killed and two or three wounded. After topping off its coal from the captured steamer, the *Vladimir* returned to Sevastopol two days later with two prizes in tow, the other being a passenger transport loaded with iron. The captured *Barwaz al-Bahr* was brought into Russian service as the *Kornilov* (after her captor) and was eventually scuttled alongside the *Vladimir* in 1855, following the Allied capture of Sevastopol.[191]

Aside from its technical significance as the first battle between steamships and, more specifically, paddle steamers, the Battle of Penderakli, as the engagement between the *Vladimir* and *Barwaz al-Bahr* is sometimes known, the outcome of the engagement was not particularly notable in the annals of single-ship duels, as one of many in which a larger, faster vessel engineered the capture of its opponent. Nonetheless, the Egyptian crew and ship had acquitted themselves well in the face of insurmountable odds, even if the possibility of exaggeration in Russian reports must be borne in mind. The next naval battle involving Egyptian warships was



Figure 4: Alexey Bogolyubov, Action between Russian steam frigate Vladimir and Ottoman-Egyptian steamer Pervaz-i Bahri of November 5, 1853, oil on canvas, 1850s.

more straightforwardly disastrous. This was the destruction of Osman Pasha's squadron at Sinope, arguably the most contentious event in Crimean War naval history. These disputes arose almost immediately after the battle and cover all its aspects, from command responsibility (for the absence of ships of the line) to charges of desertion (against the English advisor Adolphus Slade/the Mushaver Pasha), alleged conspiracies by the Anglo-French ambassadors, and even whether the attack of a squadron of frigates with ships of the line might have constituted a breach of naval honour or even a "massacre".[192] From the point of view of the Egyptian Navy, however, these debates ultimately bear little significance, given that its role at the battle was entirely in a subordinate capacity, namely, the contribution of the frigate *Dumyat (Damietta)* to Osman Pasha's formation.

Perhaps the best that can be said from an Egyptian perspective about Sinope is that the Battle of Navarino was repeated only in form, not in scope. In 1827, the Egyptian frigates *Murshid al-Jihad*, *al-Ihsaniyya*, and *Surayya* were lost at anchor, alongside three corvettes, four brigs, and two schooners, as part of a doomed Ottoman fleet ostensibly under the cover of shore batteries, which was nonetheless decisively defeated by overwhelming force. On 30 November 1853, a similar fate befell the 7 frigates, 3 corvettes and 2 steamers in the harbour of Sinope when a Russian squadron of 6 ships of the line, 2 frigates and 3 steamers entered the harbour and demanded their surrender. After considerable vacillation and with the Russian ships already in position, the Ottoman frigate *Nizamiye* opened fire, initiating an unequal engagement that lasted from one to six hours, ending in the total destruction of the Ottoman squadron, barring the steamer *Taif,* which successfully fled to Constantinople. The *Dumyat* found herself outmatched by a Russian three-decker of 120 guns and suffered the destruction of her masts, followed by the detonation of her magazines.[193] For some of the crew aboard the Egyptian frigate, the parallels with Navarino were more than theoretical: Slade records how some of them who were veterans of the earlier battle "took to their boats or jumped

overboard after discharging a few broadsides", saving "above 300 of their number." This, combined with their proficiency in swimming, allowed the Egyptian officers and men to avoid the more than 60% loss rate which befell the roughly 3,000 men of the Ottoman force.[194]

In technical terms, the Battle of Sinope is best known for the Russian use of Paixhan shell guns, recorded to have had a devastating effect on the wooden Ottoman Warships; though given the almost three-to-one advantage in broadside weight the Russians enjoyed, the result was a foregone conclusion. Ironically, the defeat proved to be a considerable political boon to the Ottoman Empire as a belligerent act in defiance of the Anglo-French fleets anchored in the Bosphorus, which, reported as a massacre, triggered much public outrage in Britain. While causality is difficult to ascertain, the engagement at the very least served as a convenient justification for the eventual Anglo-French declaration of war in March 1854.[195]

The entry of the world's two foremost naval powers reversed the naval balance in the Black Sea, and in May 1854, the remaining Imperial and Egyptian warships formed a powerful squadron and joined with the Allied fleet of Sevastopol, acting in support of the Circassian Front. No surface actions were recorded by the Egyptian fleet for the remainder of the war, though a final catastrophe befell the ship-of-the-line *Miftah al-Jihad* and the frigate *al-Buhayra* (recorded by the Illustrated London News as *Bahiré*) on the night of October 30, 1854, when they were caught in a storm between Constantinople and Varna and lost. Estimates of the numbers who perished range from 850 to more than 1,000 and included the Egyptian admiral Hasan al-Iskandarani, with the Triest Gazette further recording that 50 of the survivors were killed by Greek residents on shore.[196] As the war drew to a close in 1855-56, Egyptian forces were gradually repatriated throughout the winter. The final solemn return took place on 12 September 1856, when the remaining Egyptian troops gathered in Constantinople for their departure.[197]

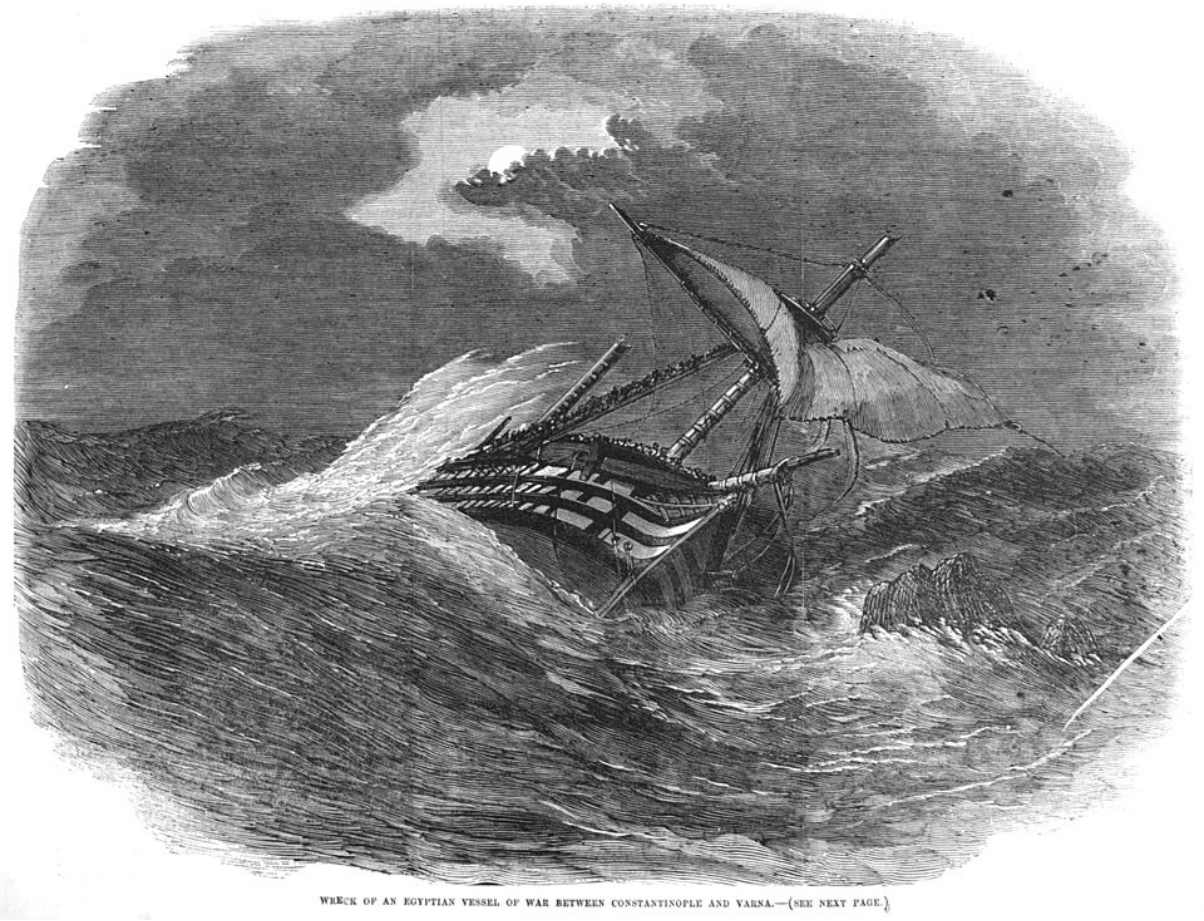


Figure 5: Wreck of an Egyptian Vessel of War between Constantinople and Varna (reproduced from The Illustrated London News, 2 December 1854).

A postscript to this conflict occurred in 1877, when renewed hostilities between Russia and the Ottoman Empire led to the despatch of a new Egyptian contingent. With tensions rising, the Egyptian fleet helped transport a convoy of 7,000 men, 40 assorted cannon, and Gatling guns. The limitations of the unarmoured Egyptian fleet were illustrated by the fact that even the Egyptian coast itself was at risk from Russian ironclads, with Ottoman escorts required to ensure safe departure. This conflict, as with the earlier 1867 Cretan war, proved largely uneventful for the Egyptian fleet, with no losses or combat recorded.[198]

Overall, the Egyptian Navy's involvement in the Crimean War and subsequent Ottoman conflicts reflected a series of political compromises between various elite factions within Egypt on the one hand, and the Ottoman Imperial government on the other. Despite its misfortunes, the fleet's officers and men had served well within the tactical and strategic constraints of the late Ottoman Navy and, moreover, had completed their core task of projecting Egyptian seapower onto land by escorting and transporting more than 40,000 men from Egypt to the front lines. Strategically, their service was not in vain, for in 1856, Sultan Abdülmecid I authorised an increase in the size of the Egyptian army by 30,000 men, laying the groundwork for Egypt's subsequent African expansion.[199]

Gunboat Diplomacy? The Egyptian Navy in the Scramble for Africa

No area epitomises the position of 19th-century Egyptian naval history at the periphery of multiple disciplines more than the brief Egyptian expansion in the Red Sea along the Horn of Africa. While located at the edges of multiple regional and imperial literatures, to date, there has been only one book-length English monograph covering the topic, Talhami's highly sympathetic *Suakin and Massawa under Egyptian rule, 1865-1885*, though Douin's multi-volume history of the reign of Ismail provides extensive chronological coverage in French.[200]

The naval relevance of Egyptian ambitions along the Red Sea and the Indian Ocean can be broken down into three thematic issues. The first is to what extent weaker, regional navies could still project power within an increasingly crowded imperial landscape. The second is the question of the Ottoman dimension: whether Egyptian expansion constituted a continuation or a replacement of Ottoman imperial rule, whether allegiance to the Sublime Porte provided any political leverage, and, finally, what impact the completion of the Suez Canal in 1869 had on the Egyptian Navy.

It is worth noting that, unlike in the Sudanese interior, the extension of Egyptian control along the Red Sea coast until as far as Zeila on the Somali Coast was not the result of direct conquest, but rather an internal territorial reassignment within the Ottoman Empire to the dynastic administration of Egypt, either from direct Imperial control or at the expense of other vassals. Ottoman control of Suakin ostensibly began as early as 1517, the same year the Mamluk Sultanate in Egypt was conquered. The city was placed under direct administration in 1554 and made the centre of the Habesh Eyalet the year after. Massawa was acquired in 1557, and the region became a base for the invasion of Ethiopia, amidst broader contestation with Portuguese forces in the Indian Ocean.[201] Thus, when Muhammad Ali first placed the region under Egyptian control, it was with the support of the then-overstretched Sultan Mahmud II, who had appointed the Wali governor of Jeddah to suppress the Wahhabi revolt. With Egypt possessing a de facto monopoly over the Red Sea and being the only force capable of countering British encroachment, the conferral of Suakin and Massawa on Ali's son Ibrahim in 1821 was largely a confirmation of a fait accompli.[202]

While Egyptian regional domination might appear assured in this period, in practice, the political geography of the region posed major maritime challenges to the projection of seapower. The primary difficulty pre-Suez Canal was the lack of major port facilities in the area – for his campaign against the Wahhabis, for instance, Muhammad Ali had to build a flotilla of 18 100-150-ton ships at the Bulaq Arsenal in Cairo and ship them in pieces on the back of camels for reassembly in Suez; to supplement this limited capacity ships had to be hired from the Sultan of Muscat.[203] Circumnavigation of Africa was an available option, undertaken by both warships and Khedivial steamships, such as those of the Aziziye Company, but involved considerable expense and difficulty.[204] The economic potential of the region was similarly not fully realised before the completion of the canal, with records showing, for instance, that the vast majority of commerce from Massawa in 1864-65 was short-distance and conducted with the ports of Aden and Jeddah. Most of the imported goods originated from India or were European re-exports, such as copper and cotton thread or linen from England.[205] The Egyptian conquest of Northern Nilotic Sudan ironically worked against this trade, as the provision of an easier route towards Egypt and its Mediterranean ports by widening the Nile cataracts and constructing ports along the Nile had a dampening impact on the caravan trade with the Red Sea, while simultaneously, political instability in the Ethiopian highlands impacted trade from the South.[206]

These limitations may explain why, although the rights to Suakin and Massawa were first ceded to the Sultan as part of the Hedjaz province in 1840, were reacquired by Muhammad Ali following

negotiations in 1846, then relapsed back to Ottoman rule upon his death in 1848, neither power was able to fully secure the broader Red Sea littoral until Khedive Ismail's renewed push of the 1860s, the area around Suakin being largely inhabited by hard-to-control pastoralists, while Massawa was periodically threatened by tribal uprisings and the forces of Tewodros II of Ethiopia.[207]

Egypt's renewed imperial push took on a far more aggressive character, benefiting as it did from several key naval developments: the renewal of regular Indian Ocean monsoons in the 1860s, the development of steamships, the completion of the Suez Canal (from 1869 onwards), and initially, the support of the increasingly influential British. Historiographically part of the high "scramble" for Africa, this period of imperial expansion saw a broad Egyptian territorial push into the African interior, often led by European and American mercenaries like



Figure 6: J. Bartholomew, Historical Map of Africa, 1885.

Werner Munzinger, Samuel Baker or Charles Chaillé-Long.[208] Within this broader context, the strategic value of the ports along the Red Sea to Egypt increased significantly, with them now serving as both key logistical bases for further expansion and a screen for the Suez Canal and Egyptian inland ambitions, in addition to their growing trade volumes.[209]

To this end, Suakin and Massawa were secured in perpetuity in 1865 following negotiations with Sultan Abdulaziz, on the dual pretext that Bedouins were fleeing taxes across provincial boundaries and that the Hedjaz authorities were too weak to prevent foreign interference by the Ethiopians and the Lazarist Missionaries. Outrage over the continued slave trade from Massawa to Jeddah was also leveraged to secure Anglo-French support on the issue, which together secured the transfer at the cost of a £E32,500 increase in Egypt's annual tribute (£E7,500 of which was to compensate the diminished Hedjaz). The preparation of the region as a staging post began almost immediately, with the initial force landed at Massawa, 800 regulars of the 3rd battalion, 10th regiment, already observed by Munzinger to be both in excess of the Port's revenues and its immediate garrisoning requirements.[210]

The early 1870s saw a spate of Egyptian territorial expansion directed along four main axes. To the South-West, the Bahr al-Ghazal and Sultanate of Darfur were conquered in alliance with the former slave trader and warlord Zubayr Rahma Mansur by 1875. To the South, upstream of the Nile, Baker's expedition (1869-73) and his successor, Charles Gordon (1874-76), annexed the regions of Equatoria. The final two directions were interlinked – the occupation of ports along the Red Sea and Indian Ocean coast, a necessary prerequisite for the inland advance into Ethiopia.[211]

The threat of European competition in East Africa was similarly becoming apparent. Having annexed Aden in 1839, the British soon reaped the benefits of their support of the Egyptian position when, in 1867, Napier's punitive invasion of Ethiopia was able to use Massawa as a base with logistical support from Egyptian Steamships.[212] In 1869, in defiance of Egyptian claims to the region, the Italian Rubattino shipping company purchased Assab Bay from the Sultan of Raheita, which prompted a forceful Egyptian response. The gunboat/corvette *al-Khartum* landed troops, and, following a brief skirmish, the Italian flag was lowered; henceforth, the area would be aggressively patrolled by Egyptian warships to establish a presence.[213]

From 1867 onwards, Egyptian influence began to extend to the Somalian Coast with the despatch of an agent to settle disputes between Somali factions. This was followed in 1869 with a visit by the Khedival steamer Sydney, and again in 1870 by a naval mission under Navy Commander-in-Chief Jamali Bey aboard the *al-Khartum,* which landed troops as a demonstration at Bulhar. Despite circumspect British opposition, these missions gained the support of the local elite, and Ottoman flags were observed flying along the Somali Coast.[214] In 1872, Keren was occupied, while Berbera was occupied by Radouan Bey in 1873 with the Corvette *al-Saiqa*, reinforced the following year by the aggressive Munzinger and the Corvette *Latif*, which anchored off the coast as a show of force.[215] Faced with this fait accompli, the British vacillated, their aversion to Egyptian expansion within Aden's sphere of influence offset by the protection that Egyptian rule offered against rival European powers, and the trade advantages it entailed courtesy of favourable treaties with the broader Ottoman Empire. Even before the question of the closure of the Bulhar trade against British protests could be fully resolved, Egyptian forces had landed at Zeila with the *al-Khartum* following negotiations with Constantinople, which settled the issue by securing imperial recognition of Egypt's rights over the Somali Coast in exchange for another increase in tribute.[216]

The final leg of Ismail's Indian Ocean expansion would be undertaken by the former British Admiral Mackillop Bey, whose career covered the First Opium War in 1842, the New Zealand War in 1846, as well as the Crimean War, and whose experience in Egypt included the post of Controller-General of Ports and Lighthouses as well as leadership of the naval academy. Selected both for his naval experience and diplomatic ability, Mackillop's mission was to take control of the coast as far as the mouth of the Juba River, territories under the tenuous control of the Sultanate of

Zanzibar. Zanzibar was a settler colony of the Omani Empire that, since that Empire's division in 1856, existed as a major regional entrepôt under British protection. It was well understood that the Sultan's forces on land and sea were no match for the Egyptians, with what little naval capacity Bargash might have possessed having been devastated in the 1872 hurricane. The scope of Ismail's ambitions can be seen in the fact that this was to be conducted in conjunction with an overland march by Gordon south towards Lake Victoria. [217]

In 1875, Mackillop departed Massawa with the frigate *Mohammad Ali* and the transports *Tantah* and *Dessouk*, joining the corvette *Latif* at Siyara, along with 5 companies of troops under the American colonel Charles Chaillé-Long and two senior Egyptian officials who were to handle local politics. The squadron first made for Ras Hafoun, just South of the tip of the Horn of Africa. From the start, the expedition was hampered by dust-filled coal and a lack of supplies, but nonetheless reached Brava without incident. There, the sailors were initially refused access to water, but this was resolved with a letter to the Emir stating that the warships were in the region to protect its Muslim tribes. At Kismayo, a garrison of 600 was overawed with a combination threat of naval bombardment and bayonet charge, and the Egyptian flag was raised in place of the red Zanzibari ensign. With a 95-man garrison left in the renamed Port Ismail, the squadron returned to Brava, disarming its 5-man guard and replacing the Zanzibari flag with the Egyptian one. A self-sufficient military settlement was established at the mouth of the Juba, and the waters now under Egyptian control were surveyed. Thus, despite an embarrassing incident in which the *Tantah* had to put in at Zanzibar for coal, receiving only a small quantity at an overpriced rate, alongside the admonitions of Sultan Bargash, the Egyptians appeared to have secured yet another fait accompli, seizing 200 miles of coastal territory at a stroke.[218]

Despite attempts to utilise pan-Islamic sentiments, Egyptian control was tenuous and dependent on fragile local alliances from the start, and any initial goodwill was soon replaced by disenchantment over "modern" restrictions and tariffs. More problematically, expectations for expansion of the slave trade under a Muslim power, as wished for by slave owners and slavers, were replaced by disappointment with its official forbidding under British pressure. The Egyptian advance was halted at Lamu, where a small force of thirty men found themselves in no position to attack the seemingly well-defended settlement, and the presence of the warships proved an insufficient threat. Though a renewed attempt was readied, by then the Sultan of Zanzibar had attained British support. Though the energetic response of the British Consul John Kirk, who threatened to bombard Brava with the corvette HMS *Thetis* after he was denied entry, attracted much attention, it was the British threat to withdraw credit that was decisive in achieving Mackillop's recall at the end of 1875.[219]

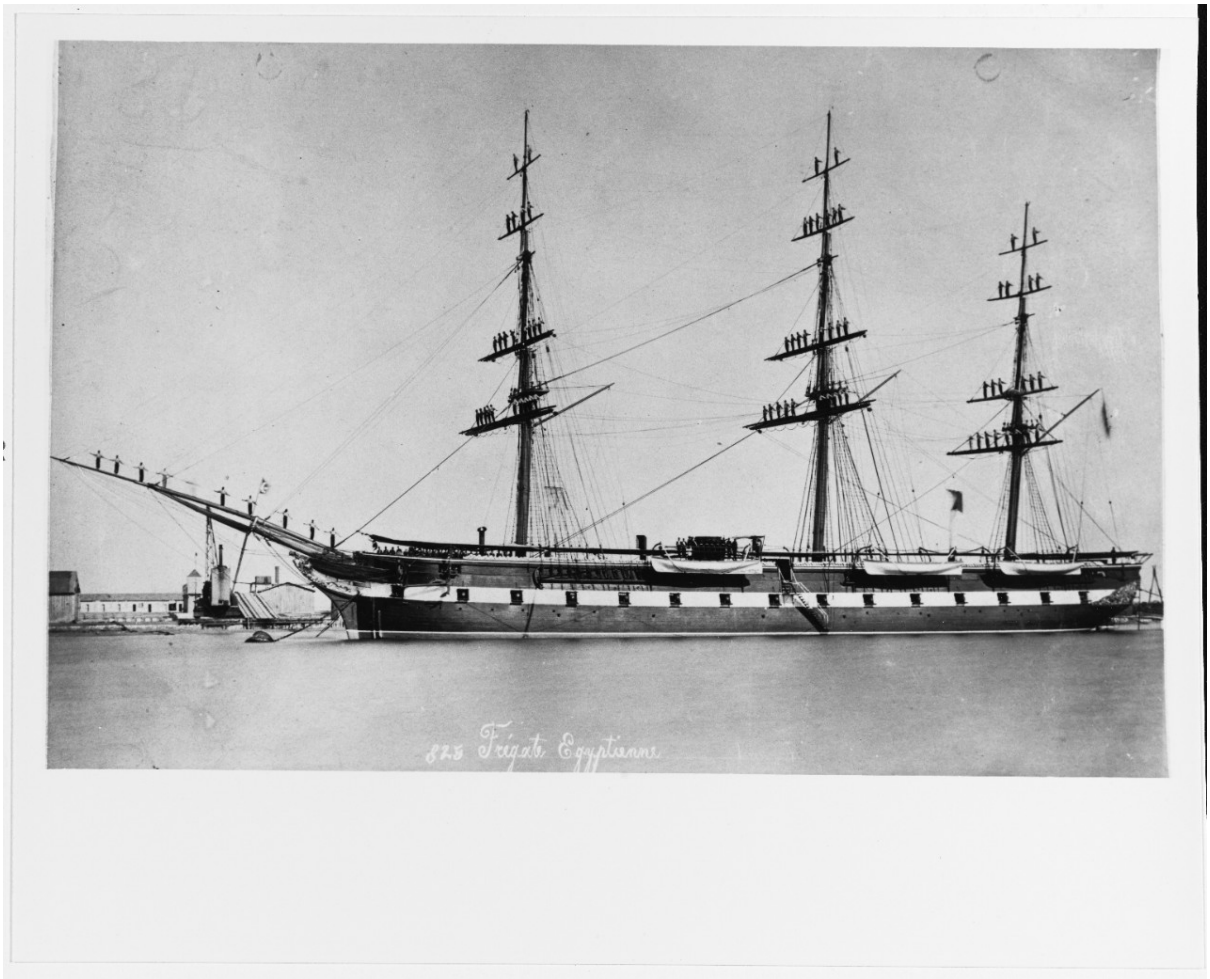


Figure 7: MEHEMET ALI (Egyptian Steam Frigate), c. 1860–1880. Naval History and Heritage Command, NH 94216

The failure of Mackillop's expedition preceded a series of military disasters that would culminate in the events of 1882. Egyptian penetrations into the Ethiopian interior resulted in costly defeats. A small column under Munzinger was ambushed and massacred by the Afar. At Gundet, a 2,500-man Egyptian force marching from Massawa was annihilated by the Ethiopian Emperor Yohannes IV, a defeat repeated the year after at Gura, when a 13,000-man force was decisively halted. The conquest of Harar by Muhammad Rauf proved to be the sole success of an expensive campaign. Egypt's accumulated debts provided a pretext for foreign intervention, and Ismail was forced to accept increasing European control until his deposition by Abdülhamid II in 1879 in favour of the more pliant Tewfik. By the 1880s, the beleaguered remnants of the positions along the Red Sea were abandoned and taken by the British, Italians, Ethiopians, and Mahdists with little resistance, and serious thought was given by both London and Constantinople to restoring Ottoman authority over Suakin.[220]

At 7 am on the 11th of July 1882, a British squadron of eight ironclads and five wooden gunvessels opened fire on the defences of Alexandria, heralding the British invasion of Egypt itself. That the pretext for the bombardment, that Egyptian forces under the authority of Ahmed Urabi refused to stop work on their fortifications, was challenged even at the time as nakedly unjust,[221] affected neither the firepower nor the protection of the British capital ships, whose shells devastated the Egyptian forts and the city behind them in a two-day battle. By the 14th, British Marines had taken control of the city, beginning a two-month campaign that would result in the formal inauguration of British colonial rule in Egypt.[222]

Conspicuous by its absence was the Egyptian Navy, which took no part in the defence of the harbour, nor factored into British pre-battle planning, so dramatic was the imbalance of power at sea. In 1884, much of what remained of the seagoing Egyptian Navy, having since been laid up in port, was sold off by a special naval committee, followed soon after by the complete abolition of the Egyptian Navy in the interests of economy, on the recommendation of Lord Northbrook. The naval undersecretary, Mustafa Pasha, and his officers were retired on pension, while other personnel and assets, such as ships, machinery and dockyards, were reassigned to other departments, such as the coast guard and the yachts and lighthouses department or sold. The Navy would only be reestablished 62 years later in 1946.[223]

Yet, though no longer an official institution, there remained a final postscript to the Navy's legacy. As the now Anglo-Egyptian forces attempted to preserve their position in Sudan from complete collapse, they called upon the services of the Gunboats on the Nile, small paddle-driven vessels rudimentarily protected from small arms fire, commanded by tribal leaders, and carrying a crew of enslaved black soldiers, Egyptian technical specialists and enslaved women who fulfilled logistical tasks such as food preparation. Overcrowded, particularly when carrying troops aboard and operating worn-out, overburdened craft, the crews of ships like the Bordein and Talahawiye nevertheless performed sortie after sortie during the ill-fated 1884-85 siege of Khartoum and the Gordon relief expedition, repeatedly running the gauntlet of Mahdist batteries until their vessels ran aground and were lost. Though gunboats would be assembled once again for Kitchener's reconquest campaign, by then, little of the previous Egyptian Navy remained, with all the officers now seconded from the Royal Navy.[224]

From a naval perspective, the Egyptian imperial project during the Scramble for Africa is revealing of the value of regional navies in the late 19th century. Despite its limited size, the Egyptian Navy was able to effectively project seapower regionally against both regional rivals like Zanzibar and speculative imperialist adventures such as those of Italy, strategically largely attaining Ismail's ambitious coastal ambitions. Even at its lowest ebb during the riverine campaigns, with the force officially disbanded, what remained of the Egyptian Navy still constituted an irreplaceable component of the Anglo-Egyptian projection of force against Mahdist forces. While superficially and with hindsight, it might appear that the great disparity in naval resources was a major cause of Egypt's inability to challenge British imperial claims, a closer reading of the successful Egyptian seizure of Berbera and Bulhar, the leveraging of financial rather than military pressure against Mackillop's expedition, and the later British acceptance of the German seizure of Zanzibar's mainland holdings despite a similar naval disparity, suggests that the core cause of Egyptian failure was one of national economic dependency rather than military weakness in-theatre.[225]

Politically, the continued Ottoman sovereignty that Egyptian Rule could claim served as an effective legitimation strategy, with the caliphal status of the Sultan used as religious leverage against newly acquired Muslim subjects, while the Porte's diplomatic sensitivity and extended historical claims provided a degree of defence against European encroachment. The perception of Turkish-ness was received as more than a mere ploy in East African memories of the Egyptian invasion, with rumours spreading that the invasion was a conspiracy between the Sultan and the "Turks" to rid the region of British influence. The employment of European officers and the adoption of a pro-trade, pro-missionary, anti-slave-trade position similarly provided some diplomatic cover, but by the mid-19th century, this was sufficiently common practice that its effects were at best limited.[226]

Conclusion

The history of the Egyptian Navy in the latter 19th century was a clear demonstration of how a second-tier, non-Western force can successfully project seapower at a strategic level. It was also a success story of how, despite political instability, the complexities of the late Ottoman world, growing Western encroachment, and an economic-industrial base that fell increasingly behind global standards, a force was consistently able to maintain its operational competence and complete the demanding tasks set to it by its political masters. Yet despite this, the force has received scant recognition in English historiography, falling entirely between multiple peripheral regional and military literatures, its significance dismissed superficially as a product of its unimpressive size.

This dissertation has corrected this gap in the literature by providing the first empirical overview of the navy of the period, through a multi-disciplinary synthesis of English-language regional literature on the Ottoman Empire, Egypt, Sudan, the Horn of Africa, the Red Sea, and the Indian Ocean; military histories of the Crimean War and Egyptian, British, and Italian imperial campaigns in Africa; as well as dedicated naval histories of the period through which hitherto lacking technical context relevant to the narrative events of Egyptian naval history has been supplied. The lack of specificity and uncertainty in secondary coverage of specific details has, moreover, been somewhat alleviated by reference to Arabic, French, Turkish, and Russian sources, as well as the memoirs of Western travellers and British and American archival materials.

From the perspective of naval history, this thesis highlights one of the field's core risks – that of conflating naval power with seapower, and, by extension, the value and historical significance of navies. Above all, the achievements of the late 19th-century Egyptian Navy highlight the hybridity by which influence from the sea has always been exercised, in a field where the technical capabilities of warships and their performance in battle at sea have conventionally been the main focus. To cite just one example, the Egyptians were able to secure the Somali coast through a combination of consistent patrolling, the landing of troops, politically prudent gestures vis-à-vis both the Porte and the issue of slavery, and a cultivated non-threatening, Europeanizing image, despite the overwhelming naval advantage enjoyed by their British adversaries. The success of the Egyptian fleet in projecting seapower through presence, escort, and logistical missions is thus an instructive case for the importance of these less-dramatic aspects of naval history, and the need for a more holistic approach to seapower, an aspect in which mainstream naval historians are still lagging behind professional naval theorists and the world's navies.[227]

From the vantage point of late Ottoman history, this dissertation demonstrates the late 19th-century Egyptian Navy's vital dual role, on the one hand securing Egyptian autonomy and intra-imperial territorial expansion, and on the other supporting the Porte's integrity during its internal and external conflicts, actions for which little recognition has been given in histories of the Ottoman Empire. In the Crimean War period alone, the Egyptian Navy checked a potential overreach of central authority in 1851, escorted and transported army contingents to the theatre, and facilitated Ottoman strategic autonomy by stiffening sea lines of communication with the Caucasus, in addition to suppressing Greek rebel forces in the Aegean. That this naval interplay between Constantinople and Cairo has so far been neglected highlights the need for works of transnational synthesis to supplement the standard national framing of military history, particularly within the global imperial context of the late-19th century.

While a work of institutional naval history, this overview of the fleet's personnel and ships is not without implications for broader social and technological history. With regards to Egyptian society, both the milder firsthand accounts of naval observers and the consistently good morale the fleet

demonstrated in service offer a counterpoint to the dystopian vision of biopolitical discipline and enforcement often associated with the Egyptian military of the era, and raise questions regarding the limits of state authority, particularly given the continued maintenance of Ottoman-legacy social inequities. In the case of technical development, this study has partially bridged the theoretical gap between the technical literature on naval technologies and political economy and institutional military histories of the Egyptian Navy, using a systemic comparative approach. The result is a re-evaluation of Egypt's relative technical naval decline as a structural outcome of specific changes in naval technology, combined with socio-economic development, superseding existing narratives that attribute these trends to the personal whims of rulers like Abbas I.

Finally, within imperial and world history, this thesis connects with a wider dialogue emphasising the importance and agency of secondary imperial powers, which were neither passive victims of Western oppression nor proto-nationalist forces of resistance, but instead violent imperial players in their own right.[228] The history of the Egyptian Navy provides a unique naval case study of how, even at the peak of the Victorian era, gunboat diplomacy was not a preserve of Western imperialists, and could be exercised no less effectively by "indigenous" competitors.

Appendix 1: List of naval vessels in the reign of Ismail[229]

Type	Name	Origin	Guns	Size (Pdr)	Material	Speed (Knots)	Engine Power (HP)	Tonnage (Tons)	Length (Feet)
Yacht	al-Mahrusa	London	8	24	Iron	16	800	3,517	447
Yacht	Misr	Toulon	6	24	Iron	12	600	3,924	376
Yacht	al-Gharbiyya	Toulon	4	12	Iron	10	500	3,600	295
Frigate	Muhammad Ali	USA	28	12	Composite	11	400	1,762	291
Frigate	Shir al-Jihad	Trieste	28	12	Wood	8	450	2,434	257
Corvette	Latif	Alexandria	6	73	Wood	10	300	1,250	220
Gunboat	al-Khartum	England	5	40	Wood	10	150	455	178
Ironclad	Dunqula	England	8	24	Armoured (4 inch)	4	70	300	149
Corvette	al-Saiqa	Alexandria	8	24	Wood	11	320	1,150	220
Gunboat	Sinar	England	7	40	Wood	8	80	478	162
Ironclad	No. 1	France	2	70	Armoured	6	60	88	141
Ironclad	No. 2	France	2	70	Armoured	6	52	177	130
Screw Cruiser	Al-Tur	England	2	15	Metal	9	140	352	197
Paddle Transport	Aswan	England	4	25	Wood	11	260	614	245
Paddle Transport	Shandi	England	4	25	Wood	11	260	614	245
Paddle Transport	Asyut	Alexandria	2	5	Wood	6	250	400	173

Type	Name	Origin	Guns	Size (Pdr)	Material	Speed (Knots)	Engine Power (HP)	Tonnage (Tons)	Length (Feet)
Screw Cruiser	al-Jafariyya	England	3	5	Iron	10	258	610	230
Screw Transport	Samanud	England	2	10	Wood	9	150	276	292
Screw Cruiser	Nur al-Huda	England	2	5	Iron	8	85	288	155
Screw Cruiser	Mukhbir	England	2	5	Iron	7	80	288	150
Screw Cruiser	Ajami	England	2	5	Iron	7	100	309	126
Sail Brig	No. 1	/	/	/	/	/	N.A.	500	107
Sail Brig	No. 2	/	/	/	/	/	N.A.	650	125
Sail Brig	No. 3	/	/	/	/	/	N.A.	950	136
Sail Brig	No. 4	/	/	/	/	/	N.A.	650	115
Sail Brig	No. 5	/	/	/	/	/	N.A.	640	110
Sail Brig	No. 6	/	/	/	/	/	N.A.	520	120
Sail Brig	No. 7	/	/	/	/	/	N.A.	2,000	189

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This work is also dedicated to its subject: those who served with the Egyptian Navy from 1848 to 1898. Their deeds deserve more recognition.

Notes

1. ‘Egyptian Defences’, *The Times* (1 June 1877), p. 4; author’s italics

2. Geoffrey Till, *Seapower: A guide for the twenty-first century* (Routledge, 2013), pp. 25-26.

3. Jeremy Black, *Rethinking military history* (Routledge, 2004), pp. 22-23.

4. Stephen Morillo, *What is military history?* (John Wiley & Sons, 2017), pp. 42-43.

5. Henry Dodwell, *The founder of modern Egypt: A study of Muhammad'Ali* (Cambridge University Press, 2011), pp. 259-265.

6. E.g. Afaf Lutfi Sayyid-Marsot, *Egypt in the reign of Muhammad Ali* (Cambridge University Press, 1984); Ghada Hashem Talhami, *Suakin and Massawa under Egyptian rule, 1865-1885* (University Press of America, 1979).

7. Ehud R. Toledano, *State and Society in Mid-nineteenth-century Egypt*, (Cambridge University Press, 2003); Khaled Fahmy, *All the Pasha's men: Mehmed Ali, his army and the making of modern Egypt*, (Cambridge University Press, 1997).

8. See Egyptianist edited volumes like Beth Baron and Jeffrey Culang, eds., *The Oxford handbook of modern Egyptian history* (Oxford University Press, 2024) for examples of the former, and Daniel A. Stolz, *The lighthouse and the observatory: Islam, science, and empire in late Ottoman Egypt* (Cambridge University Press, 2018); Elizabeth Shlala, *The late Ottoman Empire and Egypt: Hybridity, law and gender* (Routledge, 2017) for examples of the latter.

9. Letitia W. Ufford, *The Pasha: How Mehemet Ali Defied the West, 1839-1841* (McFarland, 2007).

10. For examples of Eastern Question framing, see Roger Charles Anderson, *Naval wars in the Levant, 1559-1853* (Liverpool University Press, 1952).

11. Steven Morewood, ‘Protecting the jugular vein of empire: the Suez Canal in British defence strategy, 1919-1941’, *War & Society*, 10.1 (1992), 82; Keith Neilson, ‘The Foreign Office and the defence of empire, 1919-1939’, in *Imperial Defence* (Routledge, 2007), p. 34.

12. Mark Whalan, ‘Khartoum 1885: The Passion of General Gordon’, *Studies in Church History*, 62 (2026), 294-302.

13. E.g. in Steven Serels, *The impoverishment of the African Red Sea littoral, 1640-1945* (Springer, 2018).

14. Elena Vezzadini, ‘Identity, history and power in the historiography of Sudan: some thoughts on Holt and Daly's A History of Modern Sudan’, *Canadian Journal of African Studies/La Revue canadienne des études africaines*, 46.3 (2012), 448.

15. See Marek PaweŁczak, ‘Relieved or Deceived? The Reception of the Egyptian Invasion of 1875-1876 in the Sultanate of Zanzibar and Its Subsequent Interpretations’, *Journal of World History*, 36.4 (2025); Nicholas WS. Smith, *Colonial Chaos in the Southern Red Sea* (Cambridge University Press, 2021).

16. See older literature like Reginald Coupland, *The exploitation of East Africa, 1856-1890: The slave trade and the scramble* (Hauraki Publishing, 2017), and newer works like Ralph A. Austen, ‘The 19th Century Islamic Slave Trade from East Africa (Swahili and Red Sea Coasts): A Tentative Census 1’, in *The Economics of the Indian Ocean Slave Trade in the Nineteenth Century* (Routledge, 2013); Ronald M. Lamothe, *Slaves of Fortune: Sudanese Soldiers and the River War, 1896-1898* (Boydell & Brewer Ltd, 2011); Jonathan Miran, ‘From bondage to freedom on the Red Sea coast: Manumitted slaves in Egyptian Massawa, 1873-1885’, *Slavery & Abolition*, 34.1 (2013).

17. For the initial Italian conquests, see Raymond Jonas, *The Battle of Adwa: African victory in the age of empire* (Harvard University Press, 2011).

18. A topical bibliography is available in John P. Dunn, ‘Americans in the nineteenth century Egyptian army: a selected bibliography’, *The Journal of Military History*, 70.1 (2006); John P. Dunn, *Khedive Ismail's army* (Routledge, 2004); William Best Hesseltine and Hazel Catherine Wolf, *The Blue and the Gray on the Nile* (University of Chicago Press, 1961).

19. For naval coverage in Turkish see Ali Fuat Örenç, ‘Kırım Harbi Deniz Savaşları [Naval Battles of the Crimean War]’, in *Savaştan Barışa: 150. Yıldönümünde Kırım Savaşı ve Paris Antlaşması (1853-1856). 22-23 Mayıs 2006. Bildiriler [From War to Peace: The Crimean War and the Treaty of Paris on its 150th Anniversary (22-23 May 2006). Proceedings]* (İstanbul Üniversitesi Edebiyat Fakültesi Tarih Araştırma Merkezi, 2007), while Nikolai F. Dubrovin, ed., *Materialy dlya Istorii Krymskoy Voyny i Oborony Sevastopolya [Materials for the History of the Crimean War and the*

Defense of Sevastopol], Issue 1 (Komitet po ustroystvu Sevastopol'skogo mosey, 1871) contains valuable information on the Russian perspective.

20. Jan Glete, *Navies and Nations: Warships, Navies and State Building in Europe and America, 1500-1860*, 2 vols (Almqvist & Wiksell International, 1993); see also Roger Chesneau and Eugene M. Kolesnik, eds., *Conway's All the World's Fighting Ships 1860-1905* (Mayflower Books, 1979); Edward W. Very, *Navies of the World: Giving Concise Descriptions of the Plans, Armament and Armor of the Naval Vessels of Twenty of the Principal Nations* (John Wiley & Sons, 1880); John Frederick Houghton, *The Navies of the World: 1835-1840* (Inkifingus, 2011).

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22. Crabbs, *The Writing of History in Nineteenth-Century Egypt*, pp. 139-143.

23. Yoav Di-Capua, *Gatekeepers of the Arab past: historians and history writing in twentieth-century Egypt* (University of California Press, 2009), pp. 81-85, 128-130.

24. Ibid., pp. 81, 164-167; Walid Atallah, 'Abd Al-Rahman Al-Rafi'i's Historiographical And Methodological Approach', *Bethlehem University Journal*, 10 (1991), 63-70.

25. Georges Douin, *Histoire du Règne du Khédive Ismaïl*, 3 vols (Société Royale de Géographie d'Égypte, 1933-1941), see also Georges Douin, *Les premières frégates de Mohamed Aly (1824-1827)* (Imprimerie de l'Institut français d'archéologie orientale du Caire, pour la Société royale de géographie d'Égypte, 1926); Georges-Edmond-Just Durand-Viel, *Les campagnes navales de Mohammed Aly et d'Ibrahim*, 2 vols (Imprimerie Nationale, 1935).

26. Derek Gregory, 'Scripting Egypt: Orientalism and the cultures of travel', in *Writes of passage* (Routledge, 2002), pp. 149-152.

27. Pascale Ghazaleh, 'Archives of Our Discontent: Nationalism and Historiography after 2011', in *The Oxford Handbook of Modern Egyptian History* (Oxford University Press, 2024), pp. 533-536.

28. While the title of Khedive was only officially sanctioned by the Porte in 1867, this thesis will follow Egyptian usage (which began with Muhammad Ali) for consistency

29. John Houghton, 'The Egyptian Navy of Muhammad Ali Pasha', *The Mariner's Mirror*, 105.2 (2019), 162, 166; note Houghton's system does not include ayn or hamza.

30. Houghton, 'The Egyptian Navy of Muhammad Ali Pasha', p. 179.

31. Glete, *Navies and Nations*, vol. 2, pp. 465-467.

32. Ibid., pp. 684-685.

33. Ibid., p. 466.

34. Rein Taagepera, 'Expansion and contraction patterns of large polities: Context for Russia', *International Studies Quarterly*, 41.3 (1997), 482.

35. Ayşegül Avcı, 'Successful failure: American naval engineers at the Imperial Arsenal in Constantinople 1831-1842', *Middle Eastern Studies*, 60.3 (2024), 354-355.

36. Houghton, 'The Egyptian Navy of Muhammad Ali Pasha', p. 167; the ships are respectively Murshid al-Jihad, Shir al-Jihad, Rashid, al-Buhayra, al-Zafariyya and Kafr al-Shaykh.

37. 'Sail Frigates (2nd Class, 58 Guns)' [www.shipscribe.com/marvap/212a.html, accessed 8 June 2026]; Stephen Shepard Roberts, *The Introduction of Steam Technology in the French Navy, 1818-1852* (The University of Chicago, 1976).

38. Houghton, 'The Egyptian Navy of Muhammad Ali Pasha', p. 175.

39. Derek James Overton, *Some Aspects of Induced Development in Egypt Under Muhammed Ali Pasha and Khedive Ismail*, master's thesis (Simon Fraser University, 1971), p. 55.

40. John Bowring, *Report on Egypt and Candia: Addressed to the Right Hon. Lord Viscount Palmerston, Her Majesty's Principal Secretary of State for Foreign Affairs, &c. &c. &c.* (W. Clowes and sons, 1840), p. 55; Houghton, 'The Egyptian Navy of Muhammad Ali Pasha', p. 175.

41. MacDougall, Philip, *Islamic seapower during the age of fighting sail* (Boydell and Brewer, 2017), p. 179

42. Fahmy, *All the Pasha's men*, p. 56.

43. Georges Douin, ed., *Une mission militaire française auprès de Mohamed Aly: correspondance des généraux Belliard et Boyer* (l'Institut français d'archéologie orientale pour la Société royale de géographie d'Égypte, 1923), p. 111, translated

by Overton, ‘Some Aspects of Induced Development’, p. 51.

44. MacDougall, *Islamic Seapower*, p. 179.

45. Ismail Sarhank, *Haqaiq al-Akhbar an Duwal al-Bihar [True Narrations of Maritime Nations]*, 2 vols (Matbaah al-Amiriyyah, 1312-1314 A.H. [1895-1898]), vol. 2, p. 340.

46. Also known as Warrior or Guerriere/La Guerriere.

47. Christopher Montague Woodhouse, *The Battle of Navarino* (Hoddler and Stoughton, 1965), pp. 116-120.

48. Umar Tusun, *Umara al-Bihar fi al-Ustul al-Misri fi al-Nisf al-Awwal min al-Qarn al-Tasi Ashar [Admirals of the Sea in the Egyptian Fleet in the First Half of the Nineteenth Century]* (Hindawi Foundation, 2020), pp. 22-23. The mouth of the Ambracian Gulf is ironically where, in 31 BC, the navy of Ptolemaic Egypt was decisively defeated by that of Octavian at the battle of Actium, an event usually marking the end of both the Hellenistic age and, a year later, the end of the ancient Egyptian polity.

49. Tusun, *Umara al-Bihar*, p. 67. The Gulf of Spada is where, in 1940, the Australian light cruiser HMAS Sydney, with 5 British destroyers, caught and destroyed the Italian light cruiser Bartolomeo Colleoni, exactly 4 months before her famous engagement with the Kormoran.

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51. Abd al-Rahman al-Rafii, *Asr Muhammad Ali [The Era of Muhammad Ali]* (Matbaat al-Fikrah, 1930), p. 321.

52. Fahmy, *All the Pasha's men*, pp. 40-43; Dodwell, *The founder of modern Egypt*, p. 39; Sayyid-Marsot, *Egypt in the reign of Muhammad Ali*, pp. 22, 196-197.

53. Fahmy, *All the Pasha's men*, pp. 25, 284, 304, 310; Toledano, *State and Society in Mid-nineteenth-century Egypt*, pp. 16-17; Jonathan Endelman, ‘In the shadow of Empire: States in an Ottoman system’, *Social Science History*, 42.4 (2018), 818; Stanford Jay Shaw and Ezel Kural Shaw, *History of the Ottoman Empire and Modern Turkey: Reform, revolution, and republic: the rise of modern Turkey, 1808-1975*, 2 vols (Cambridge University Press, 1976), vol. 2, p. 18.

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57. Houghton, ‘The Egyptian Navy of Muhammad Ali Pasha’, p. 174; Samir Ahmeti and Gazmend S. Rizaj, ‘European impact on Egyptian industrialization during the rule of Muhammad Ali Pasha (1805-1848)’, *The Journal of European Economic History (JEEH)*, 53.2 (2024), 10.

58. Very, *Navies of the World*, p. 16; James Heyworth-Dunne, *An introduction to the history of education in modern Egypt* (Routledge, 2019), p. 140.

59. Daniel Panzac, ‘Flottes et arsenaux dans l’affrontement turco-égyptien (1830-1840)’, *Cahiers de la Méditerranée*, 84 (2012), 375.

60. Fahmy, *All the Pasha's men*, p. 50.

61. Anderson, *Naval wars in the Levant*, pp. 550-551; Sayyid-Marsot, *Egypt in the reign of Muhammad Ali*, pp. 230-231.

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63. Dilara Dal, *The modernization of the Ottoman Navy during the reign of Sultan Abdülaziz (1861-1876)*, PhD dissertation (University of Birmingham, 2016), p. 28; Bernd Langensiepen and Ahmet Guleryuz, *The Ottoman Steam Navy, 1828-1923*, ed. and trans. James Cooper (Conway Maritime Press, 1995), p. 3; the Turkish authors of the latter reference book mistakenly identify one ship of the line as a frigate, giving the figure of 8 ships of the line and 12 frigates.

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66. Anderson, *Naval wars in the Levant*, p. 556.

67. Fahmy, *All the Pasha's men*, p. 289.

68. Ufford, *The Pasha*, p. 194.

69. Anderson, *Naval wars in the Levant*, p. 552; Umar Tusun, *Al-Jaysh al-Misri al-Barri wa-al-Bahri [Egypt's Armed Forces on Land and Sea]*, Safahat min Tarikh Misr [Pages from the History of Egypt], vol. 3 (Maktabat Madbouli, 1996), p. 207; Houghton, 'The Egyptian Navy of Muhammad Ali Pasha', p. 168.

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221. Best summarized by MP Henry Richard's memorable quote: I find a man prowling about my house with obviously felonious purposes. I hasten to get locks and bars, and to barricade my windows. He says that is an insult and threat to him, and he batters down my doors, and declares that he does so only as an act of strict self-defence. H. Richard, 'Supply—Forces In The Mediterranean (Vote of Credit)', *Parliamentary Debates (Hansard)*, House of Commons, 25 July 1882, col. 1778.

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